GNSS NavStar Tool User Guide

UG01-NavStarTool-C01 V1.2

2024/8/13

Contents

1. About this Guide

The purpose of this guide is to describe how to use GNSS NavStarTool.

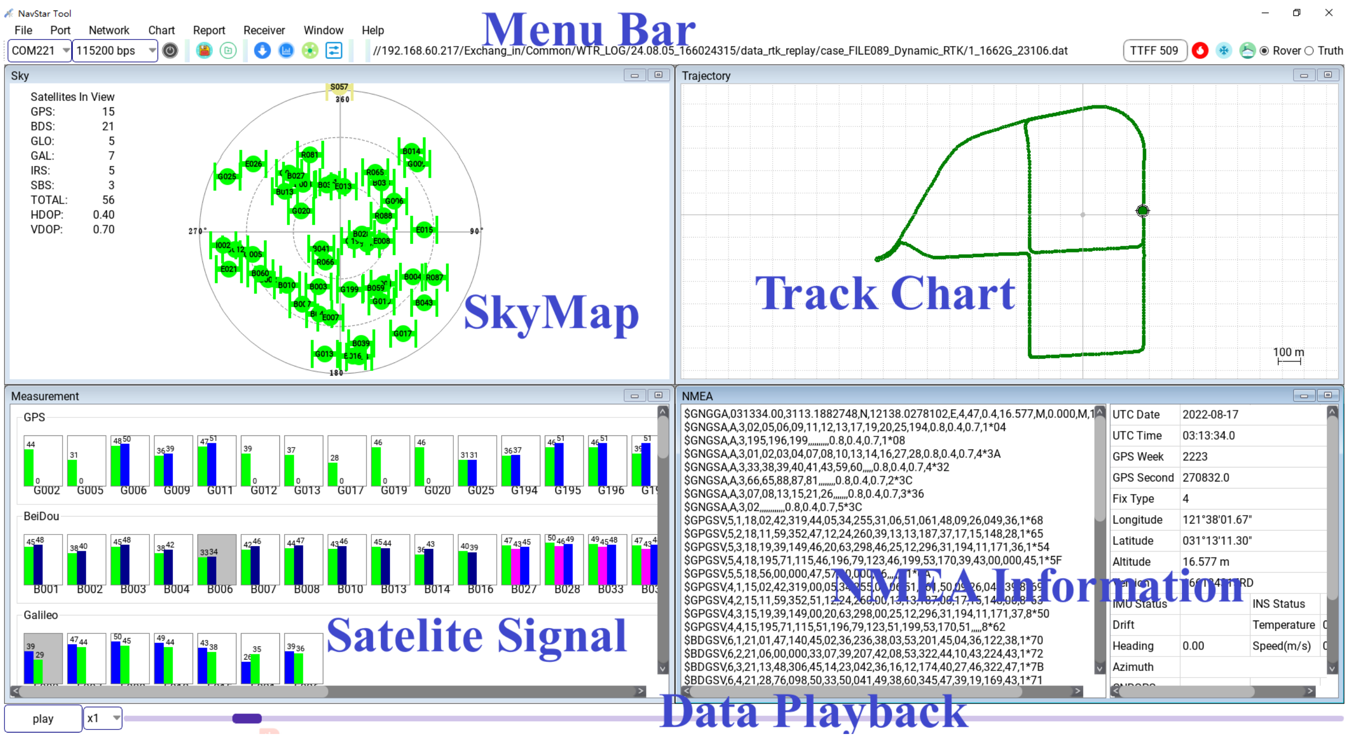
GNSS NavStarTool is an integrated software tool for the use, testing and debugging of BK GNSS chip products. It allows users to evaluate the navigation and positioning performance of BK GNSS chip products or similar modular products. Its main functions are as follows:

- **Chapter 2 Describes the tool interface.**
- **Chapter 3 Quick Operation Guide**, including recording GNSS output logs, downloading and updating the GNSS firmware, RTK Test NTRIP configuration, AGNSS manual delivery test, TTFF test, AGNSS test, RTK test, positioning accuracy analysis, etc.;
- Chapter 4-7 Other functions of the tool are introduced;
- Chapter 8 Appendix: Special file format description, tool runtime dependency solution, version description;

This document provides guidance on how to use these features and gives common use cases.

2. Interface Introduction

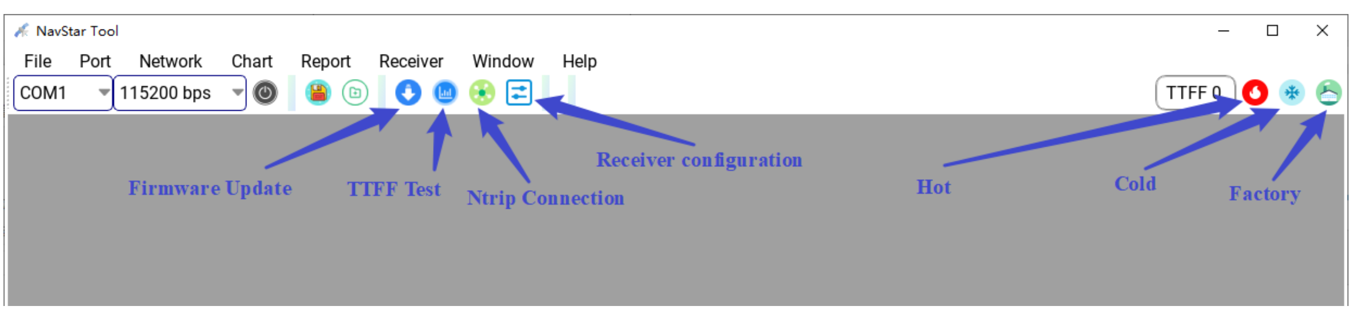
The main interface of GNSS NavStarTool is as follows:



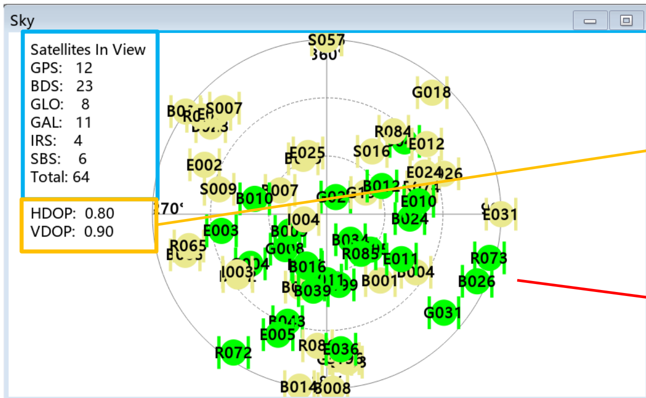
The main window information display consists of four parts, each of which supports full screen display and reduction, which can display the output information of GNSS products more conveniently.

2.1 Menu bar

As shown below, the menu bar is divided into two parts. One is the top menu bar; The second is the shortcut key for some functions.



2.2 Sky Map



Visual satellite number statistics: support GPS/ BDS/ GLONASS/GALILEO/ IRNSS/ SBAS system and their sum.

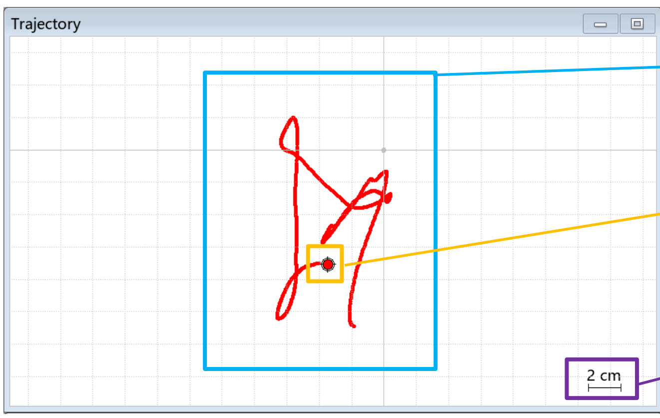
Satellite distribution configuration Mass: Horizontal Precision Factor HDOP and Vertical Precision Factor VDOP.

Sky Map: Displays the satellite's azimuth and altitude Angle information.

Yellow indicates satellites that are not involved in positioning calculation, green indicates satellites that are involved in the positioning solution.

G-GPS,B-BDS,R-GLO,E-GAL,I-IRS,S-SBS

2.3 Track Chart

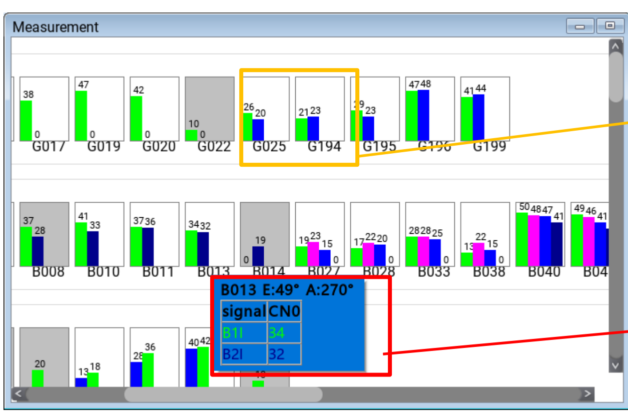


Location trace display: The tool automatically saves the latest 1000 positions Point trajectories.

Current output anchor point display: The point circled by the circular cursor is the current epoch the positioning result of the output. **Red -SPP**, **Blue -SBAS or RTD**, **Yellow-Float**, **Green -FIX**, **Pink -DR**

Scale: The scale of the positioning track display.

2.4 Satellite Signal Strength



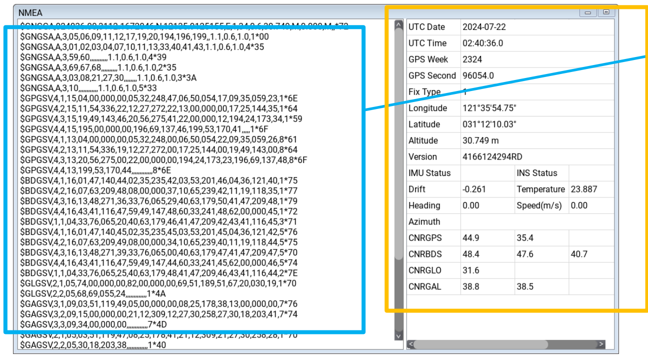
Subgraph: Shows the signal strength at different frequency under a satellite, where no the same frequency are shown in the subgraph by bars of different colors.

When the background color is gray, it means that the current satellite is not involved in the positioning solution.

Value display: When the mouse is hovering over the subgraph, the difference within the subgraph will be displayed frequency signal strength value.

Note: BK166X series, BDS B2i and B2b show the same color, PRN1~16 for B2i, PRN17~PRN63 for B2b

2.5 NMEA information



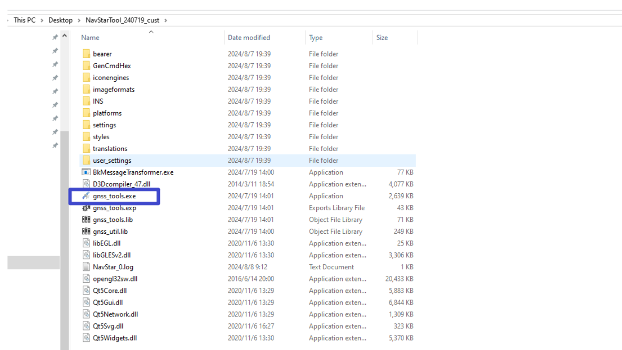
NMEA logs are displayed in real time: GGA/ GSA/ GSV/ RMC statements are supported.
Value display: Displays key location information (including firmware version information).

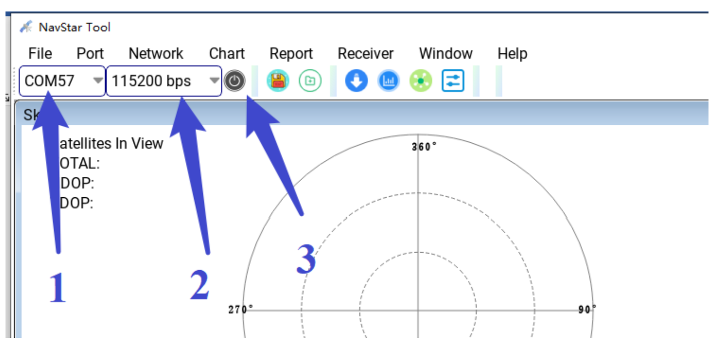
3. Quick Operation Guide

3.1 Connect the Device

a) **Preparations for Connection**: NavStar Tool, GNSS module, USB-To-Serial port board;

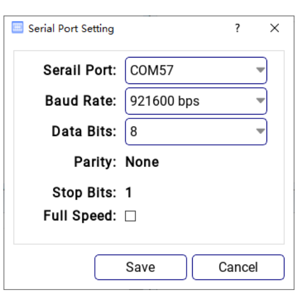
b) **Connecting Device**: Connect the GNSS module to the USB-To-Serial port board and PC, open the NavStar Tool, select the serial port, set the baud rate, and open the serial port, the operations are as follows;





Instructions:

➢ If you need to change the serial Port configuration, click the menu bar [Port] -> [Config] to modify the serial port configuration.

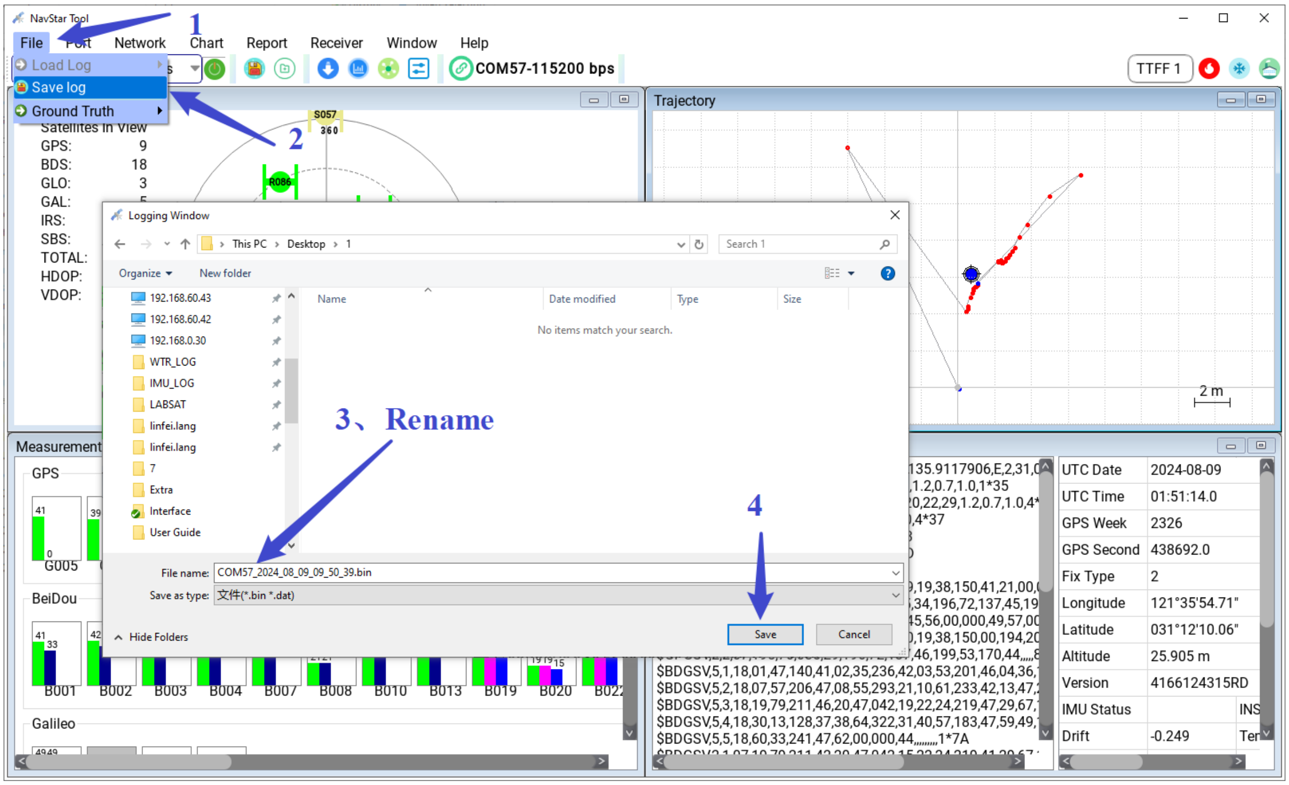


➢ **Baud Rate** Supports the Custom baud rate. Select custom and enter an integer. Do not contain non-numeric characters.

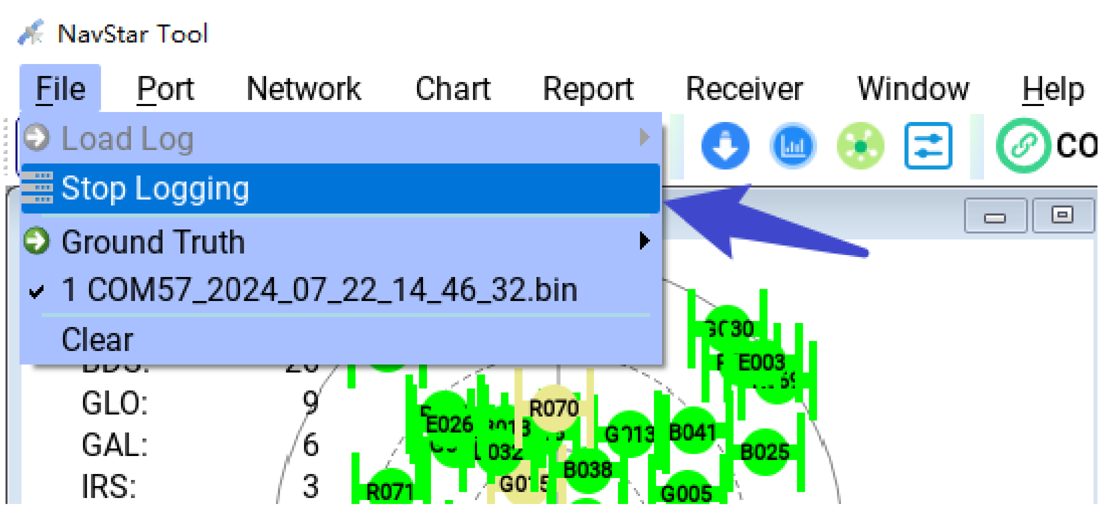
➢ **Full Speed is off by default.** After this parameter is selected, the CPU usage greatly increases to ensure the serial port communication bandwidth. This parameter is only used in R&D and debugging scenarios.

3.2 Recording Logs

a) To start recording: click the menu bar [File] -> [Save log] or click the shortcut key " ", Select the path and edit the file name;

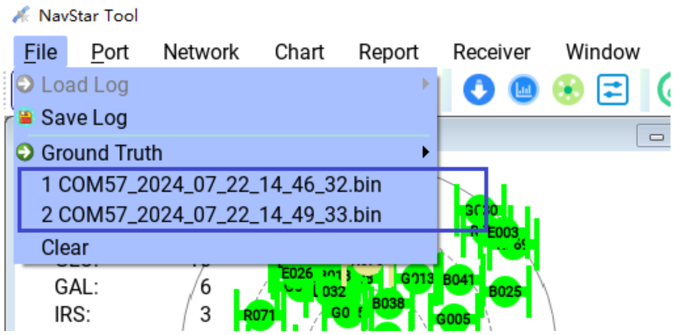


b) Stop logging: Click on the menu bar [File] -> [Stop logging] or click on " " Shortcut key;



Instructions:

The [File] submenu automatically saves the historical log file name and path.



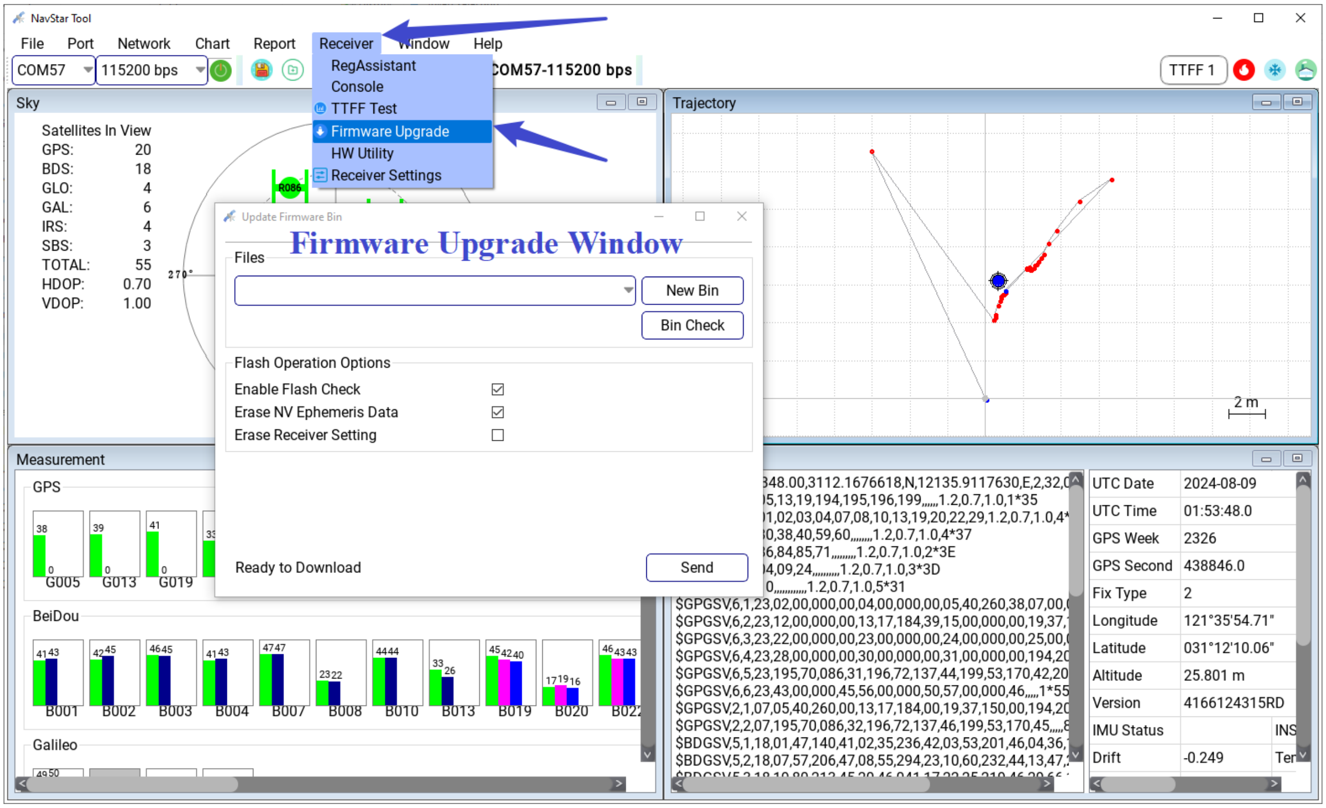
3.3 Firmware Upgrade

There are two upgrades available for the BK166X series as follows:

a) **The online upgrade** requires that the module is in normal working state. that is, the PC or MCU side can obtain correct data (such as GGA, RMC, etc.), to ensure the normal communication between the PC and the chip, the baud rate of the standard release version is generally 115200, if the baud rate is not 115200, please adjust to the correct baud rate before upgrading

b) **Reset or Power-off upgrade** can be applied to the first chip firmware upgrade or a chip upgrade that recovers after an abnormal interruption;

Tap the menu bar [Receiver] -> [Firmware Update] or shortcut key " " on the menu bar to access the firmware upgrade window.



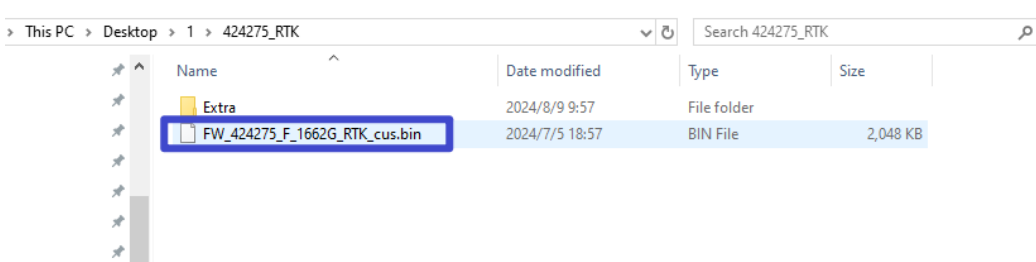
Instructions:

- ➢ If you click [Bin Check] button to check whether the firmware file is tampered with, it can be used as the check before the upgrade;
- ➢ If [Enable Flash Check] is selected, the data written into the Flash is checked;
- ➢ If [Erase NV Ephemeris Data] is selected, the ephemeris data stored in the Flash is erased;
- ➢ If [Erase Receiver Setting] is selected, old configuration items in the Flash will be erased.

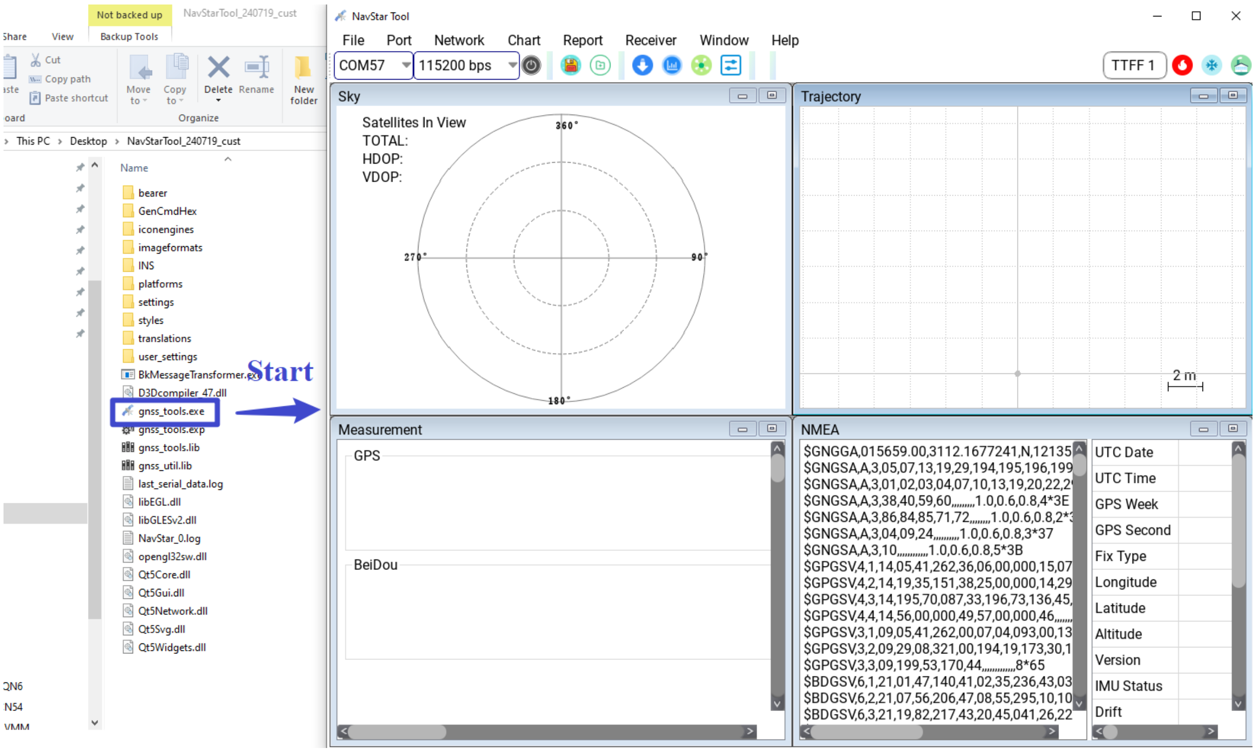
For details about the upgrade process as follows:

3.3.1 Preparations for the Upgrade

a) Firmware version preparation

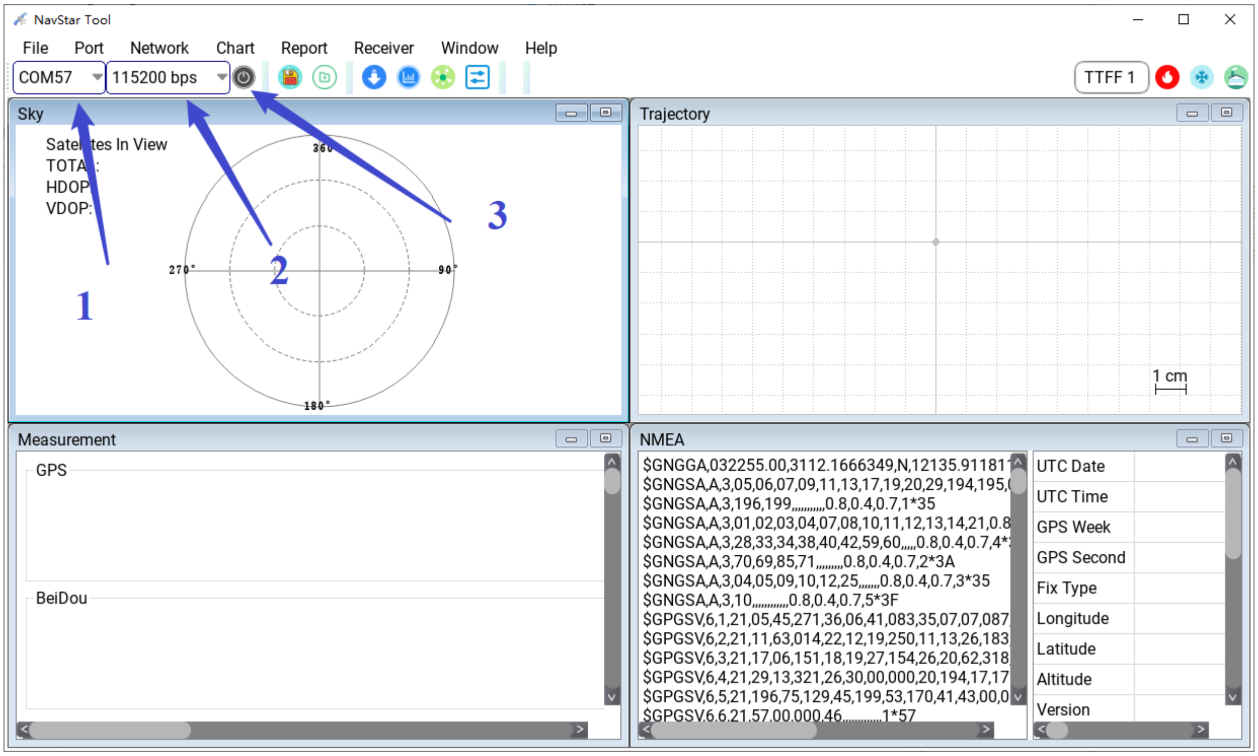


b) Upgrade tool preparation

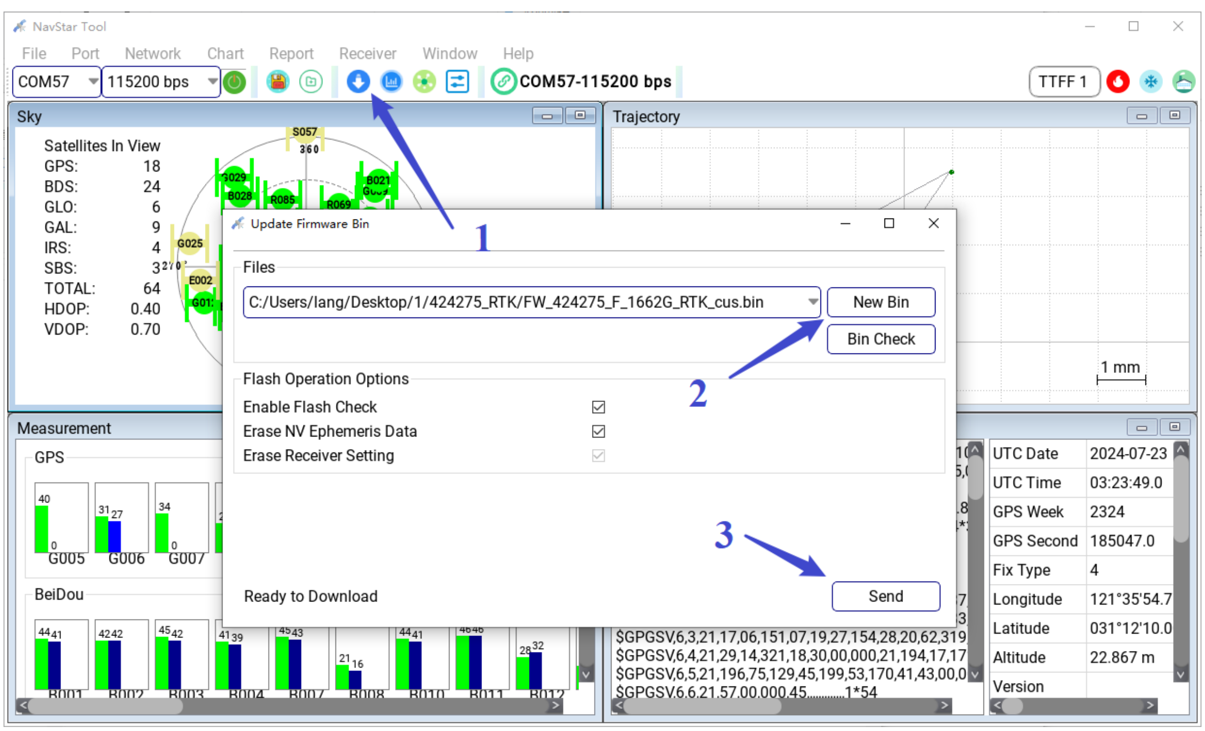


3.3.2 Online Upgrade

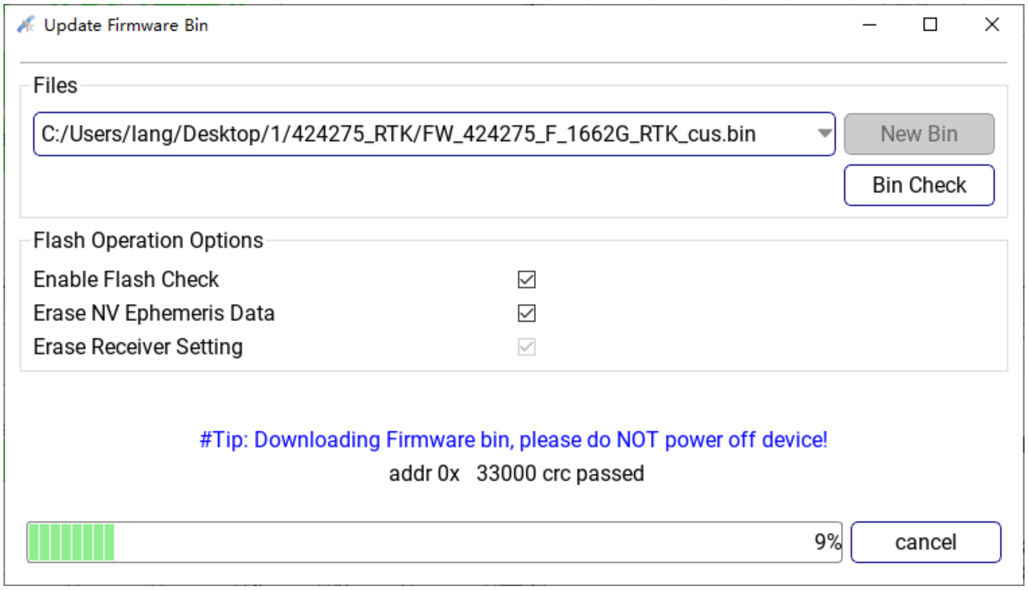
a) Open the tool and connect the serial port (Pay attention to check the baud rate, the chip needs to work properly);



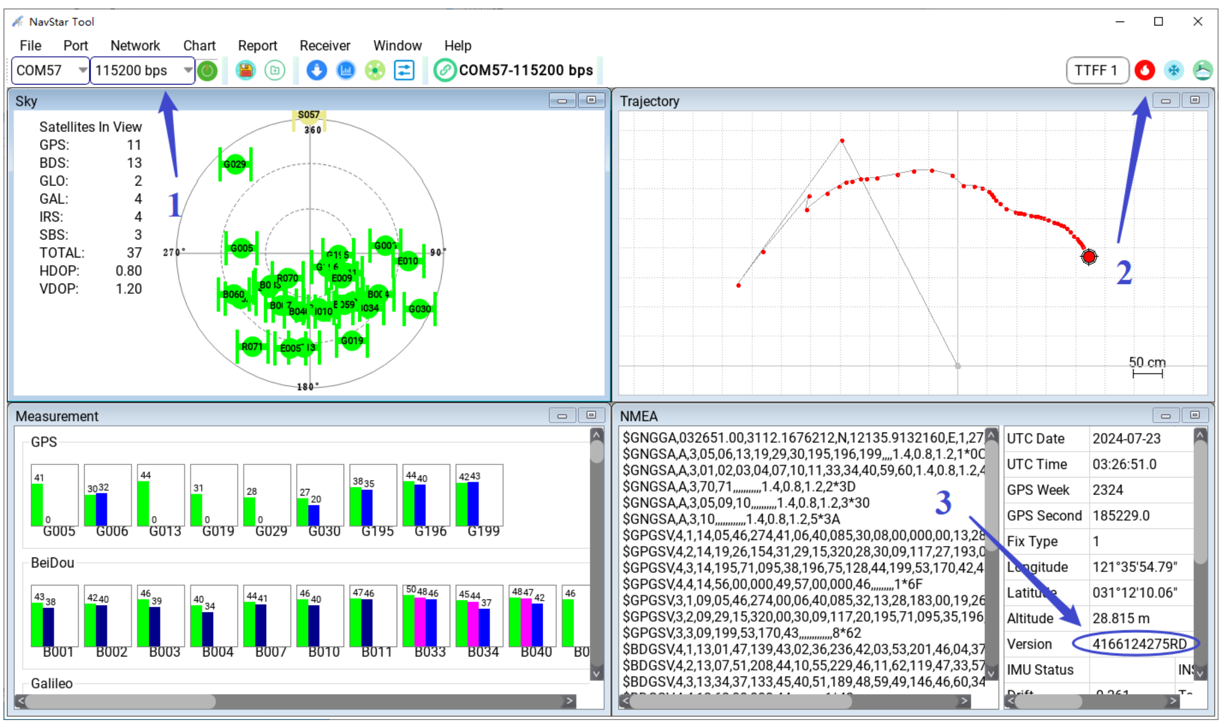
b) Open the firmware upgrade window, select firmware [New Bin] and click [send] to start the upgrade, as shown in the picture below:



c) Wait for the progress bar to indicate the completion of upgrade, as shown in the picture below:

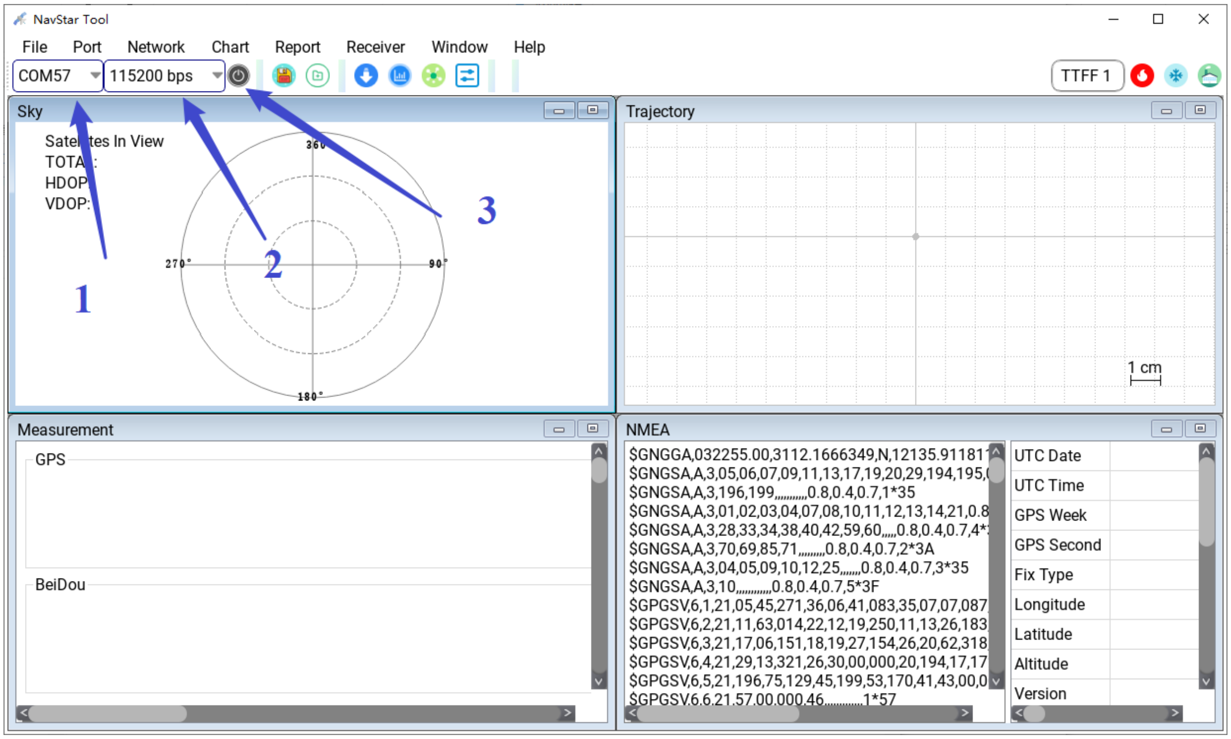


d) After the upgrade is complete, adjust the baud rate and check the version information;

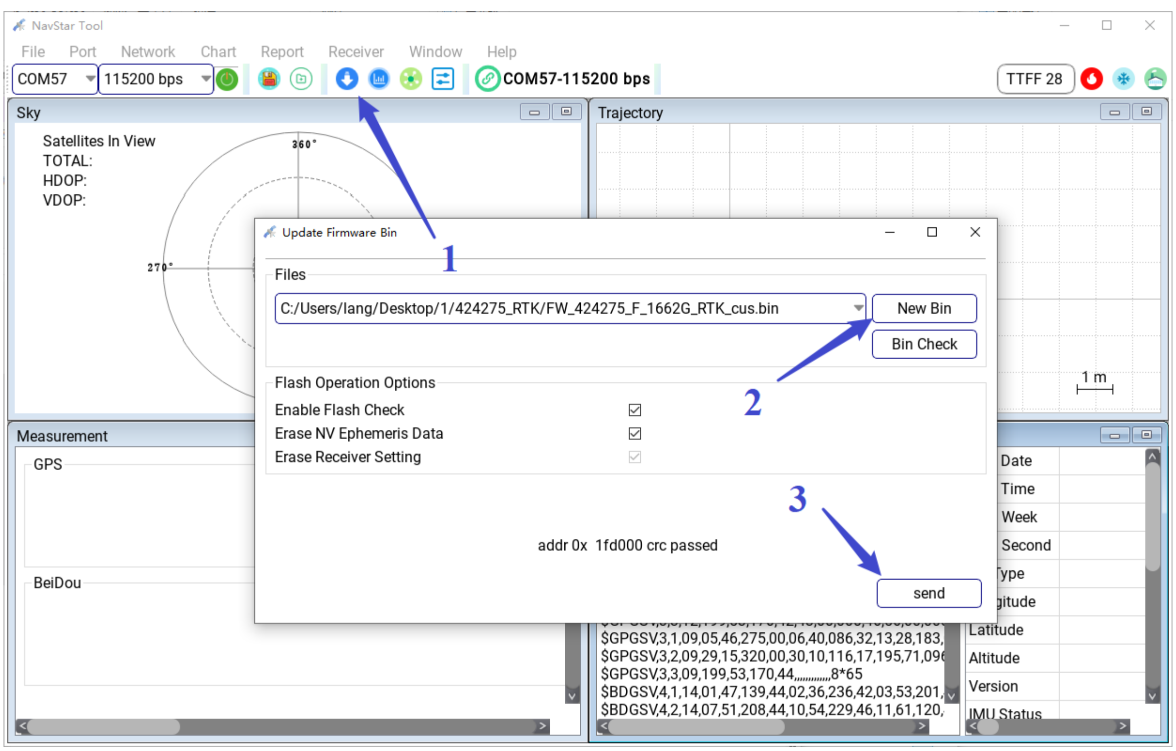


3.3.3 Reset or Power-off upgrade

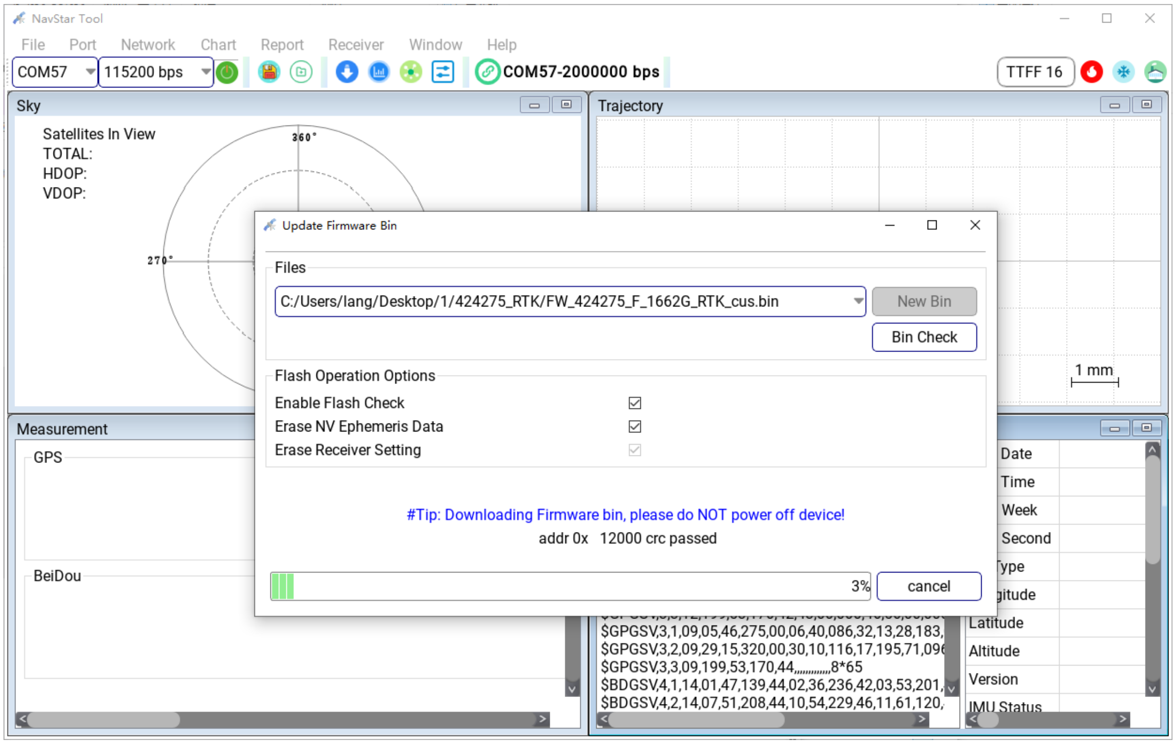
a) Open the tool, connect the serial port, **note** that the baud rate here is selected **115200**;



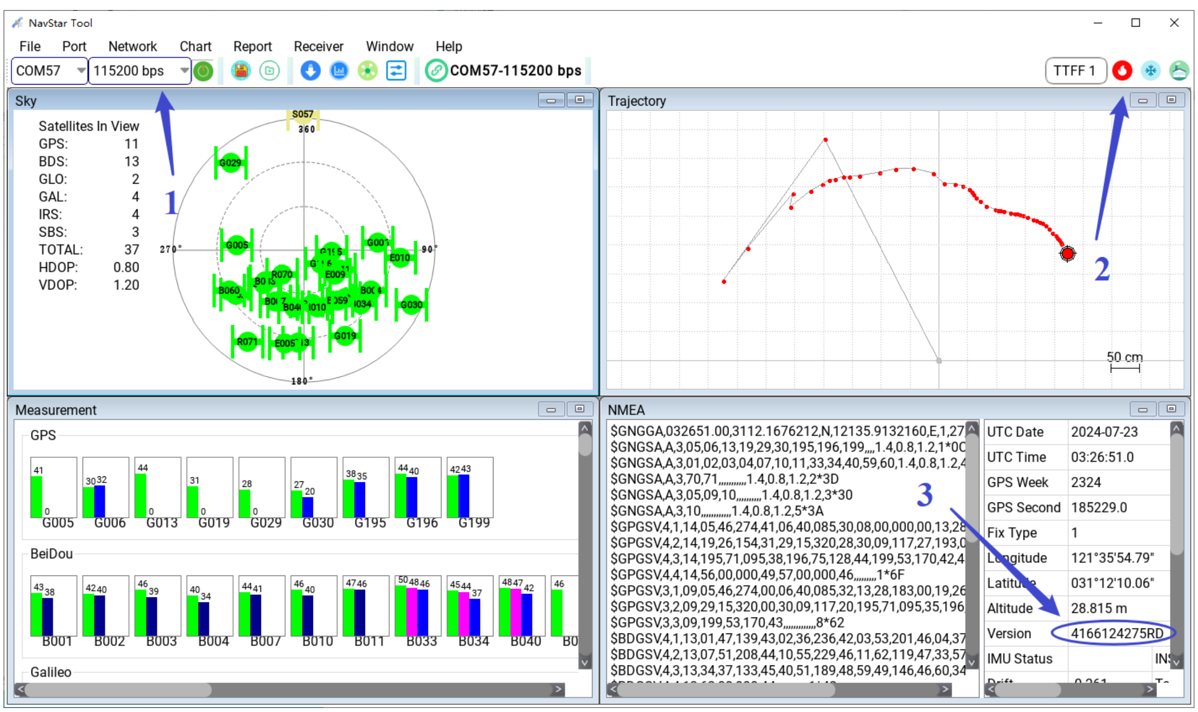
b) Open the firmware upgrade window, select firmware [New Bin] and click [send] to start the upgrade, as shown in the picture below:



c) The module is powered off, and then powered on at a interval of 1s;



d) After the upgrade is complete, adjust the baud rate and check the version information;



3.3.4 Abnormal Answer

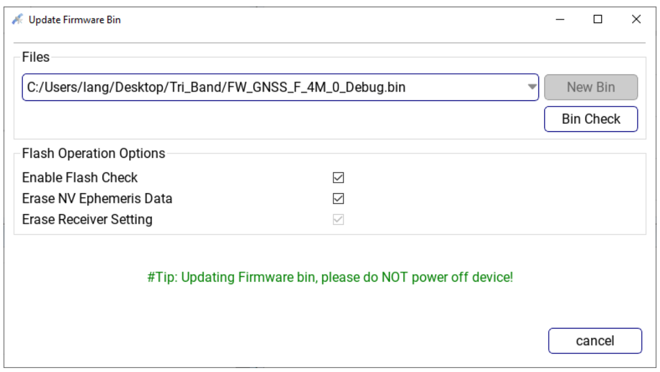
3.3.4.1 The communication between the chip and the PC is abnormal

The two firmware update methods show that the download window has no prompt and no progress bar appears. There are the following possibilities:

a) The serial port TX or RX cable is incorrectly connected;

b) The baud rate of the serial port is incorrectly selected;

c) Chip model and firmware version do not match;

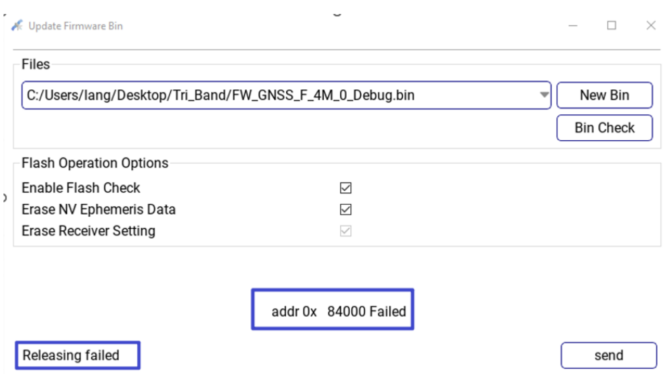


3.3.4.2 The Upgrade is Interrupted Abnormally

Two firmware update methods: "Releasing failed" and "addr 0x ***** Failed" is displayed on the download window, or the progress bar is not updated for a long time. There are the following possibilities:

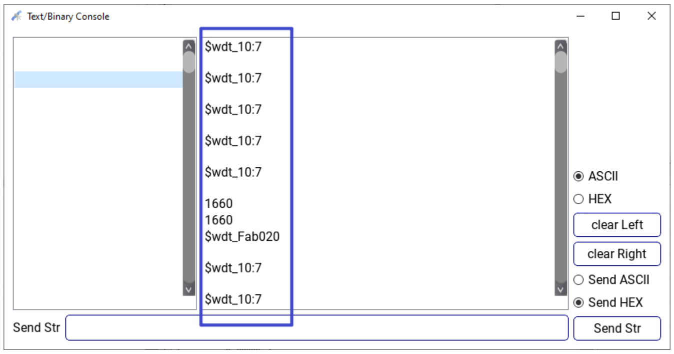
a) The serial cable is loose during the upgrade;

b) The chip is reset or powered off during the upgrade;



3.3.4.3 Chip Fails to Work After Upgrade

a) Chip model and firmware version do not match, as shown below;



b) The version configuration is inconsistent with the original configuration, the new firmware does not output the original firmware message data, and it is mistaken that the chip does not work properly;

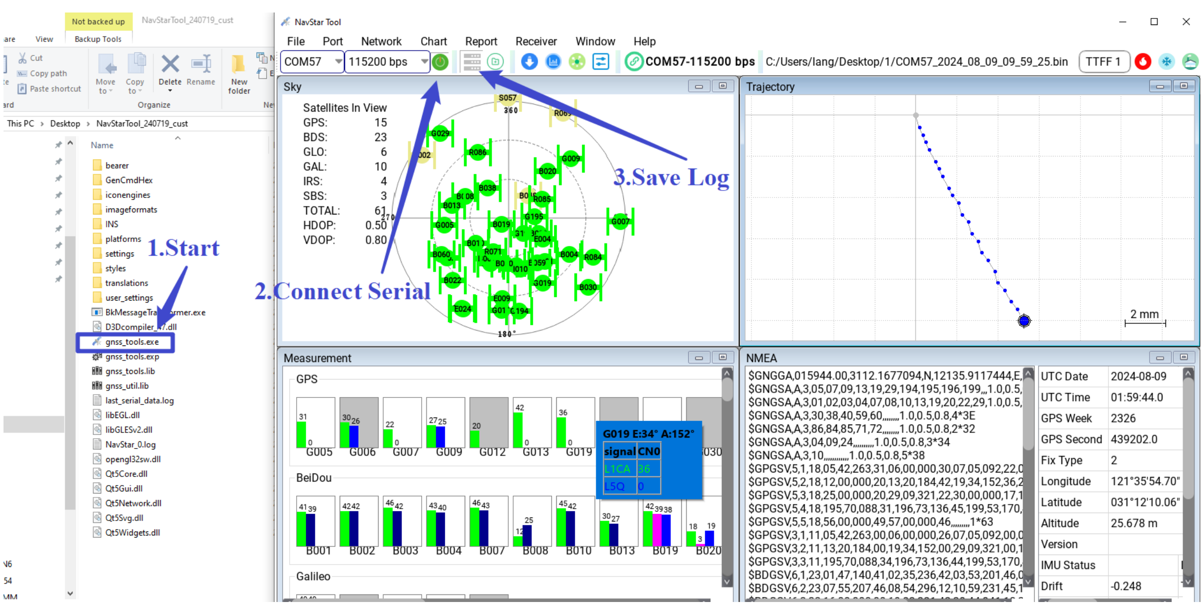
3.3.4.4 Upgrade Process Is Slow

The two firmware update methods show that the progress bar update is very slow, and there are several possibilities:

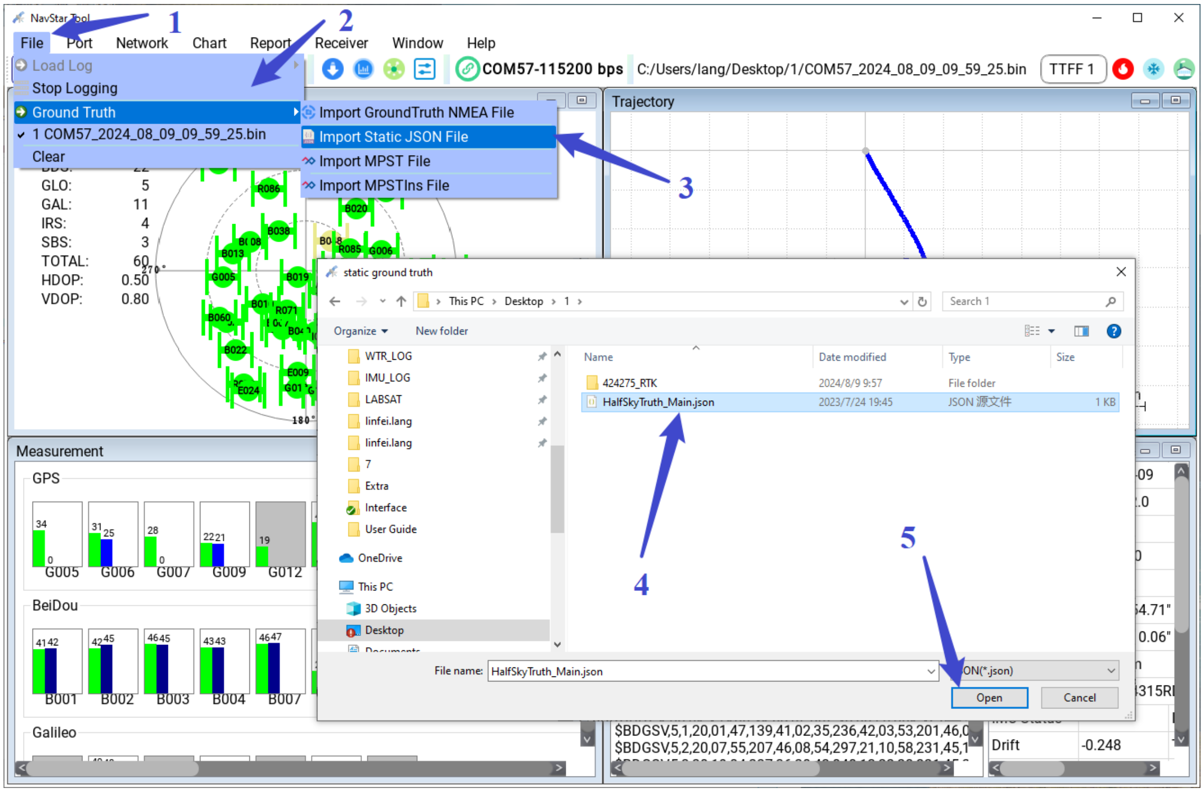
a) Serial port does not support high baud rate upgrade, usually the tool automatically increases the baud rate of the chip and tool to 2Mbps for firmware update, if the baud rate fails to increase, the upgrade will be very slow;

3.4 TTFF Test

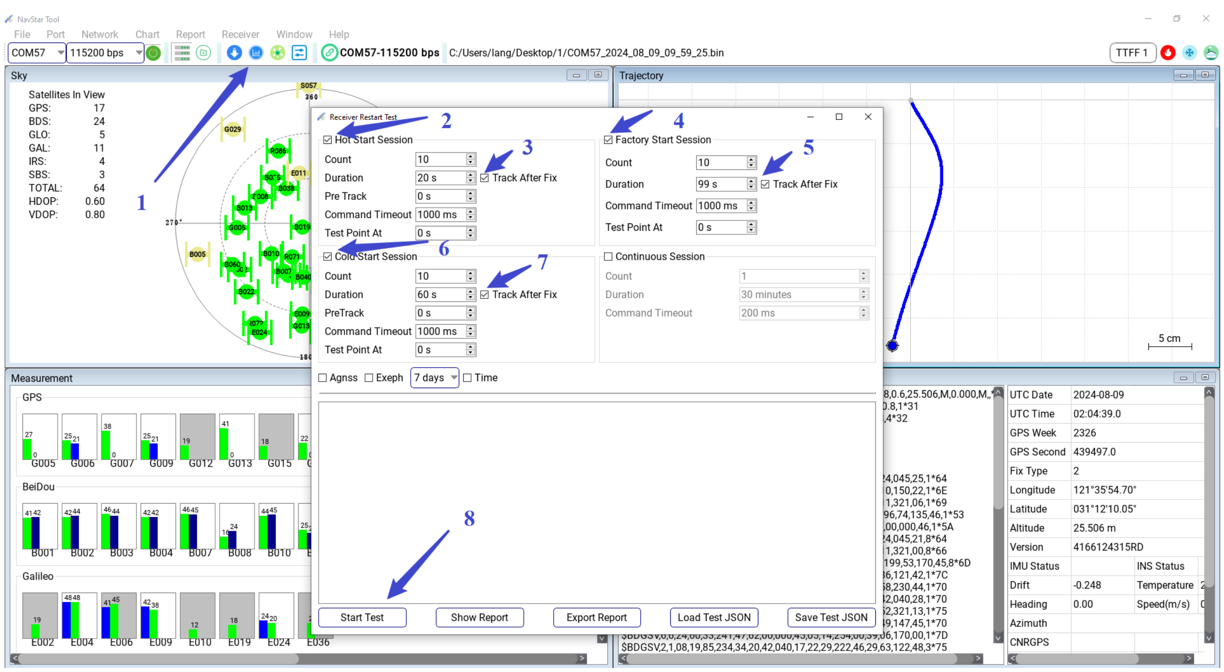
a) Open NavStarTool, connect the serial port (Refer to section 3.1), and save log;



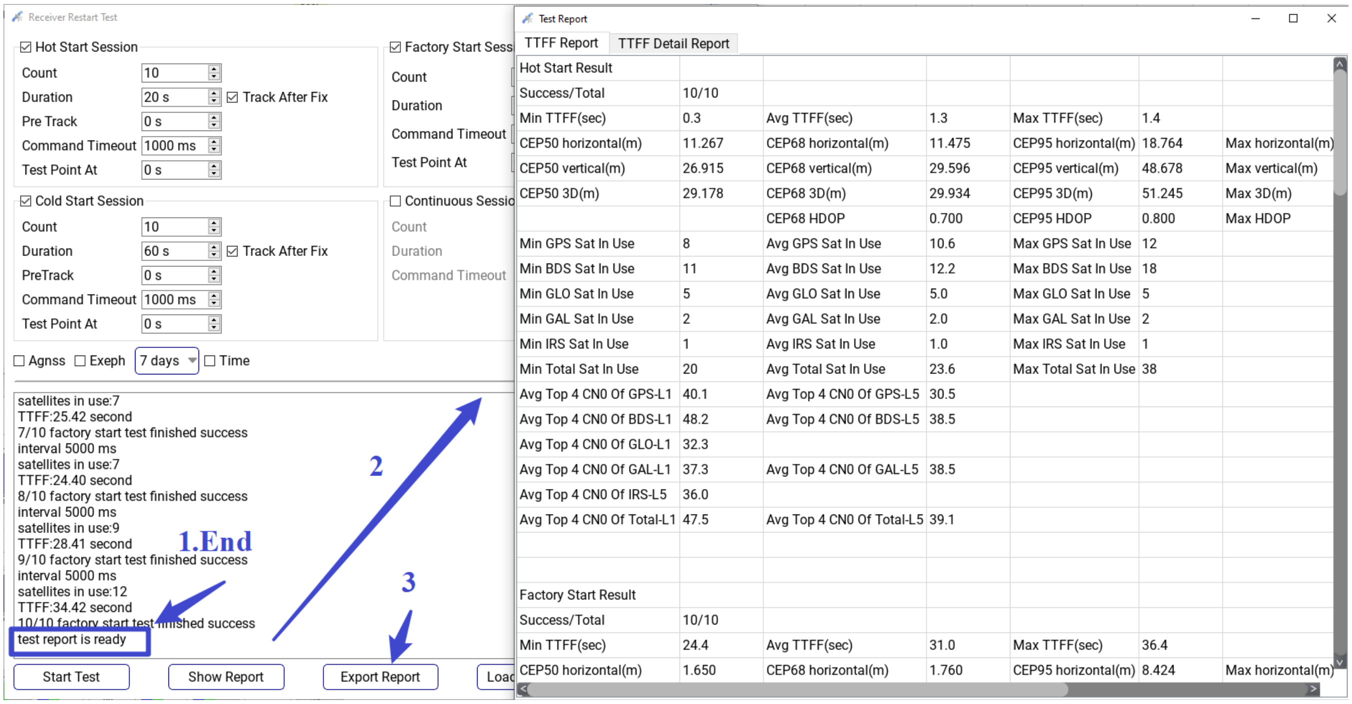
b) Import the truth file, the format is shown in Section 8.1.The JSON File Format of Ground Truth (If the truth file is not imported, the result conforms to the accuracy of the statistics);



c) Open the TTFF test window, configure the test parameters, start the test, and wait for the end of the test;

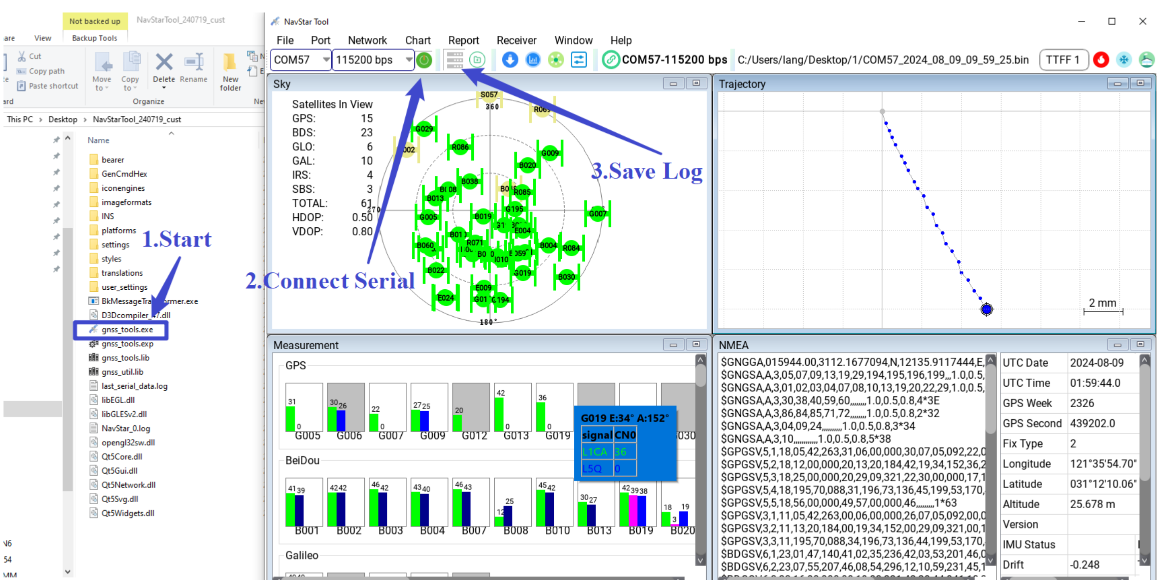


d) After the test, show the report and export the report;

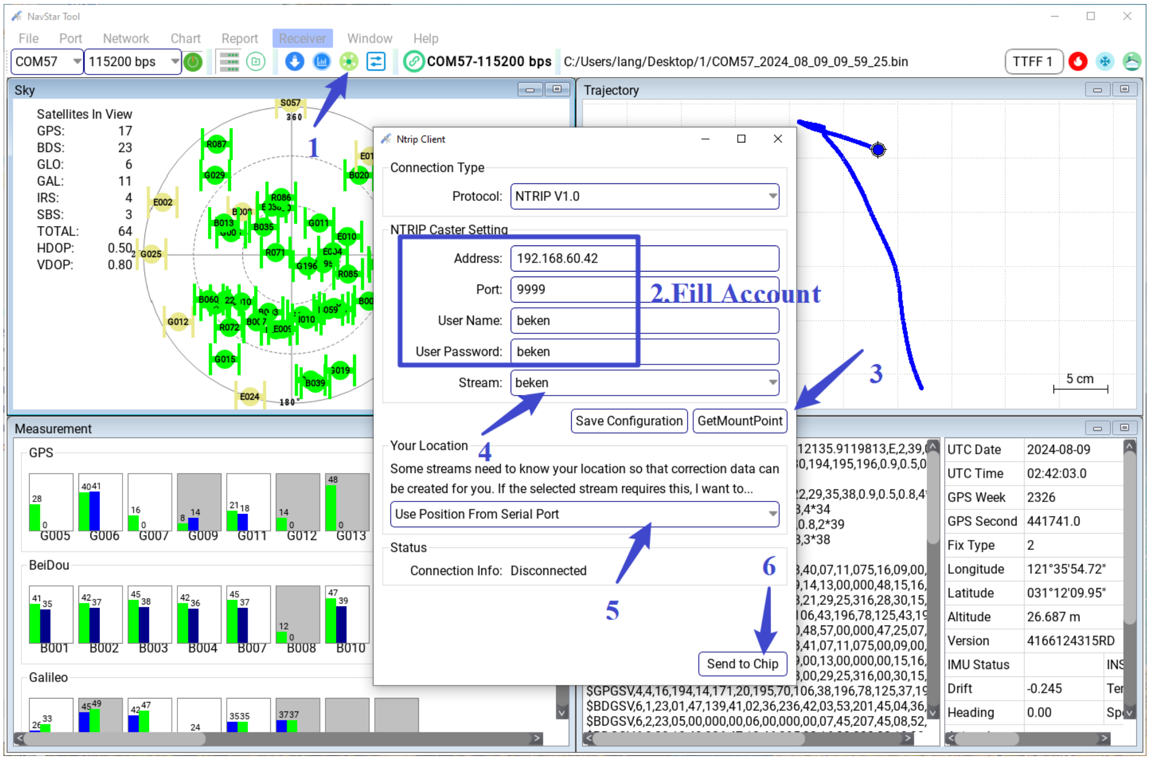


3.5 RTK Test

a) Open NavStarTool, Connect the serial port (Refer to section 3.1), and save log;

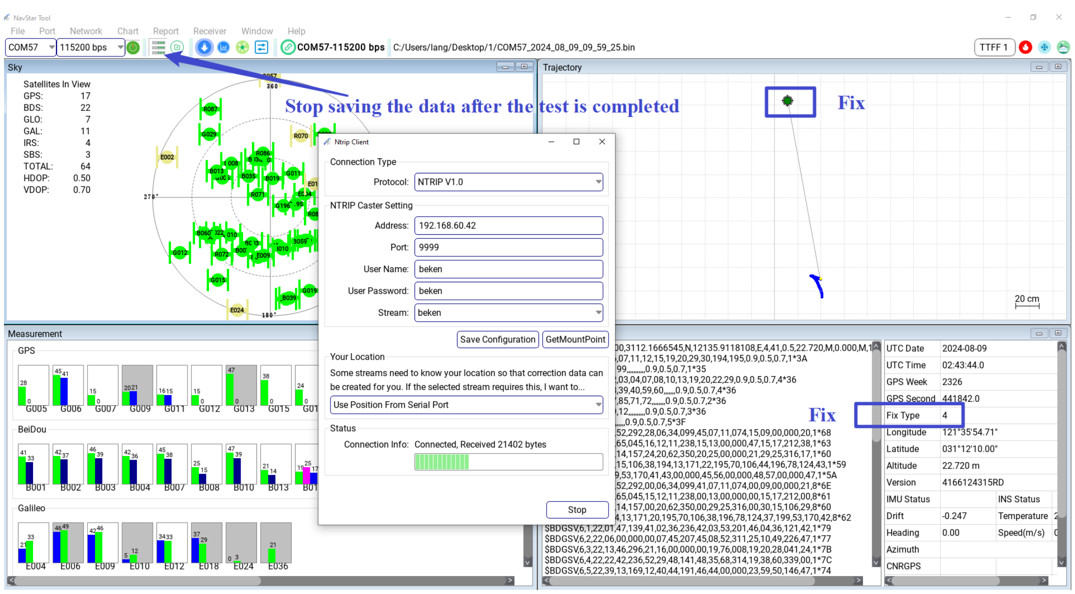


b) Click the menu bar [Network] -> [NTRIP] or the shortcut key of " ", you can enter the NTRIP configuration window and configure the RTK account, connect service;



Instructions:
Ntrip location back there are two ways to choose, one is to read the Serial Port GGA back [Use Position From Serial Port], the other is to manually enter a coordinate back [Use a manual location] .

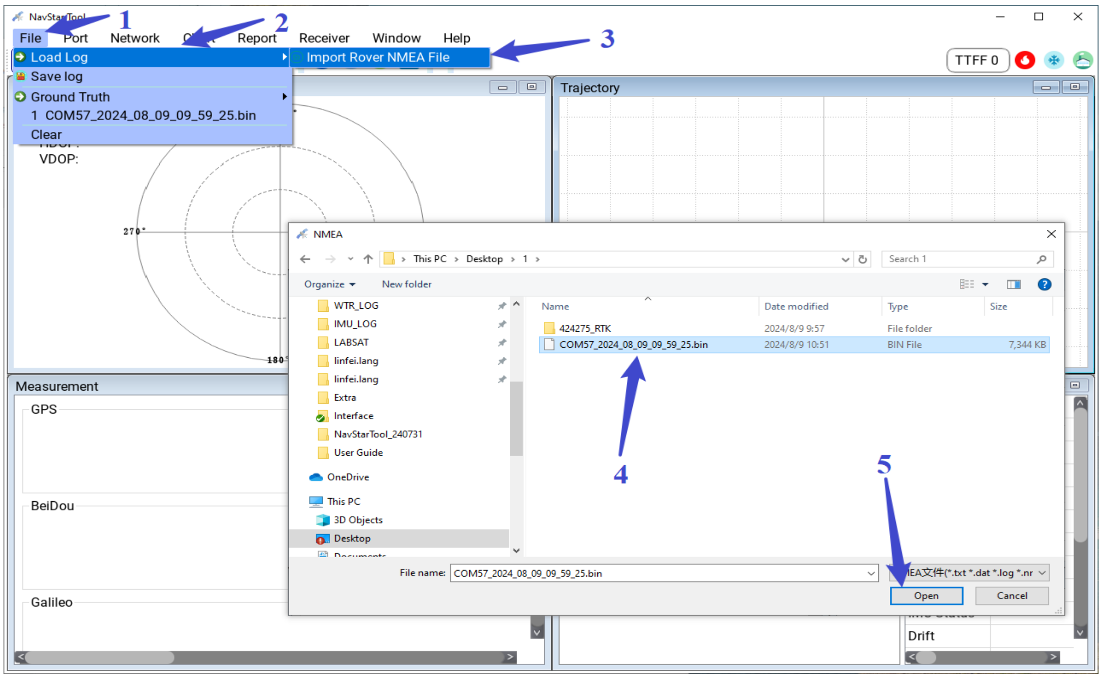
c) Check the positioning result, start the test, and stop saving the data after the test is completed;



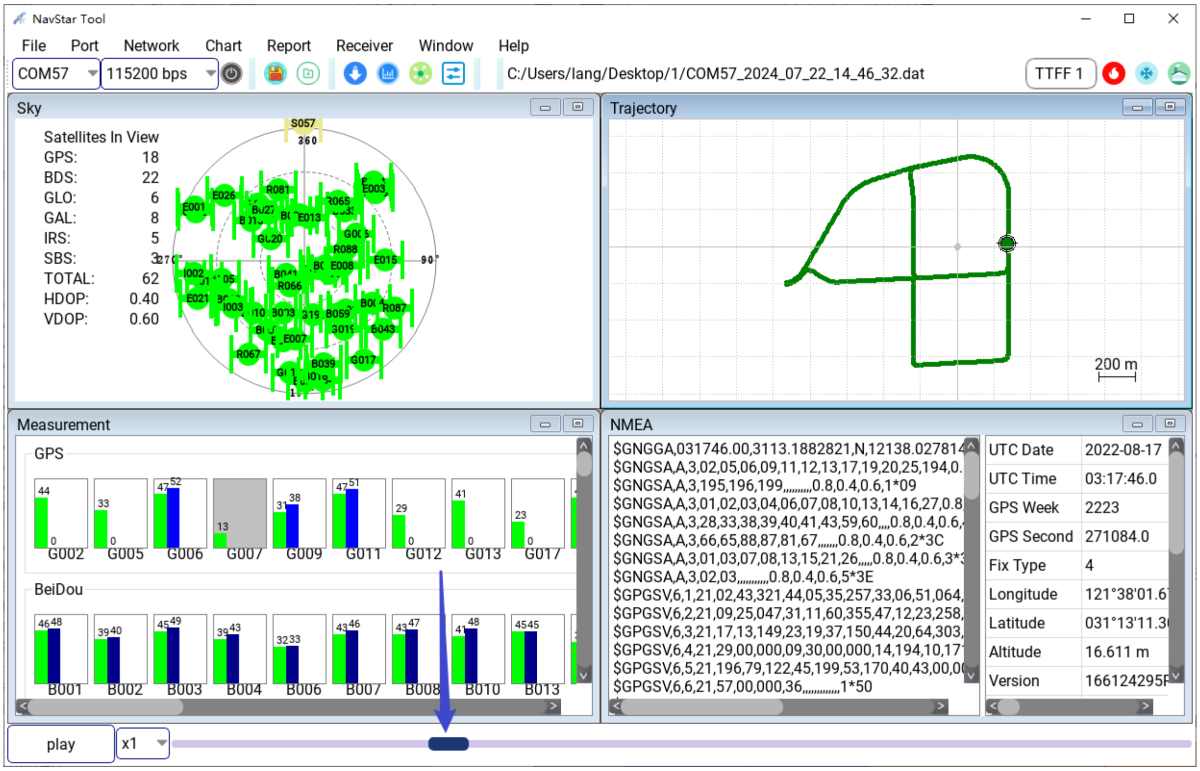
d) Positioning accuracy analysis and reporting methods are described in Section 3.9 Positioning Accuracy Analysis.

3.6 Data Playback

a) Click [File] -> [Load Log] -> [Import Rover NMEA File], the File selection window will pop up, select the NMEA file to be played back; In addition, the tool also supports File drag and drop function, directly select the NMEA File to be played back, hold down the left mouse button, and drag the file into the main window of the tool;



b) After loading NMEA File, the playback button and drag progress bar will be displayed at the bottom of the main window of the tool, and the tool can automatically play back and adjust the playback speed; You can also pull the progress bar through the mouse to locate the moment that needs to be played back;



Instructions:

➢ If click [Play] button for automatic playback of NMEA data;

➢ If you check [x1] to change the rate of NMEA data playback;

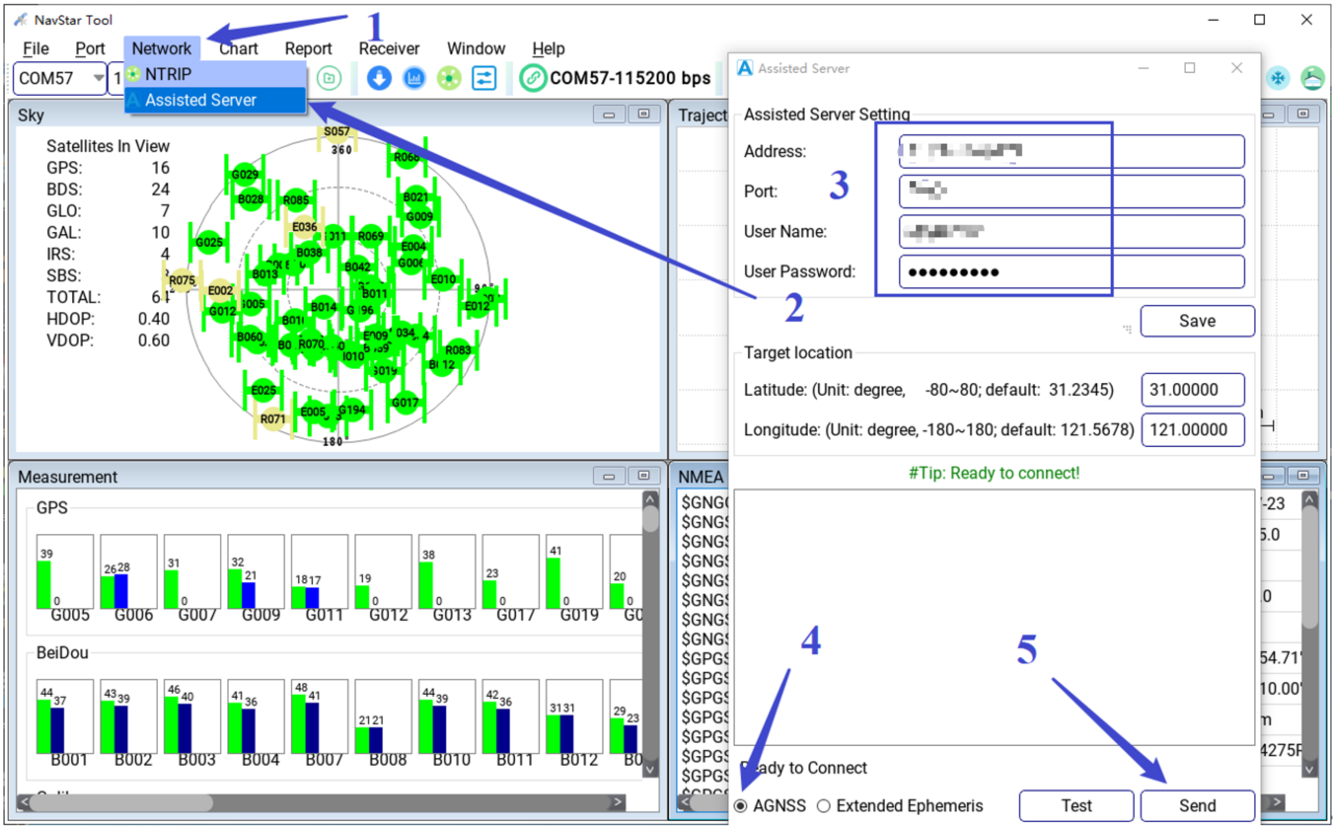
➢ The button on the progress bar can be pulled by the mouse.

3.7 AGNSS Manual Test

NavStarTool Test the BK166X series AGNSS as follows:

a) Connect the serial port. Refer to Section 3.1.

b) Click the menu bar [Network] -> [NTRIP] to enter the assisted server window, configure the AGNSS account, click [Send] to enter the ephemeris, as shown below:

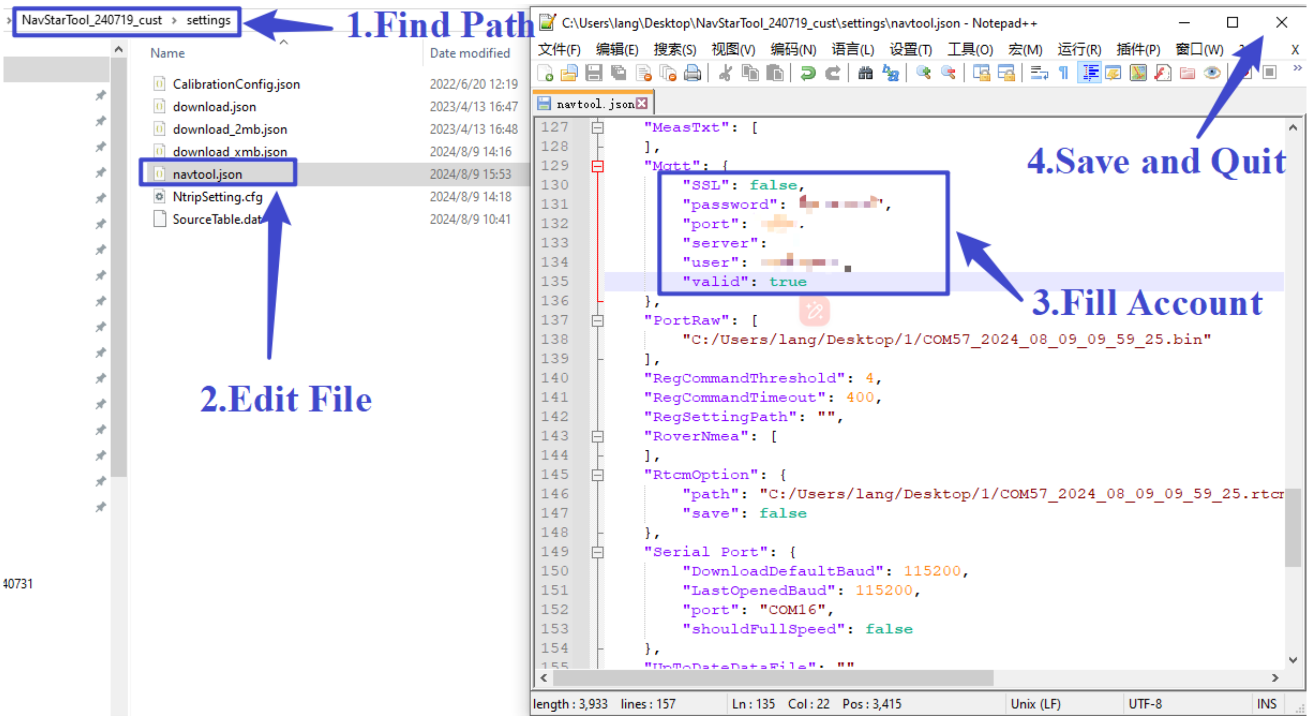


Instructions:

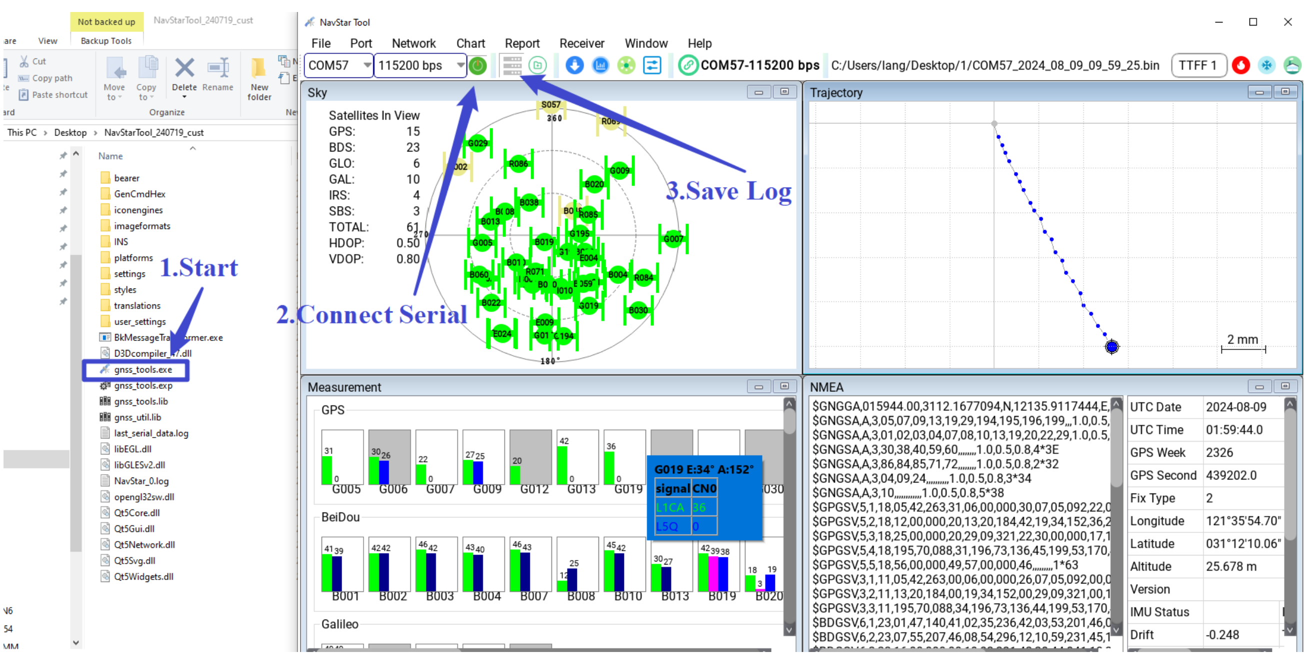
- Real-Time Ephemeris and Extended Ephemeris can be switched between [AGNSS] and [Extended Ephemeris] on the assisted server window;
- [Test] is used to test whether the account connection and data delivery are normal. To inject the ephemeris to the device, click [Send] .

3.8 AGNSS TTFF Test

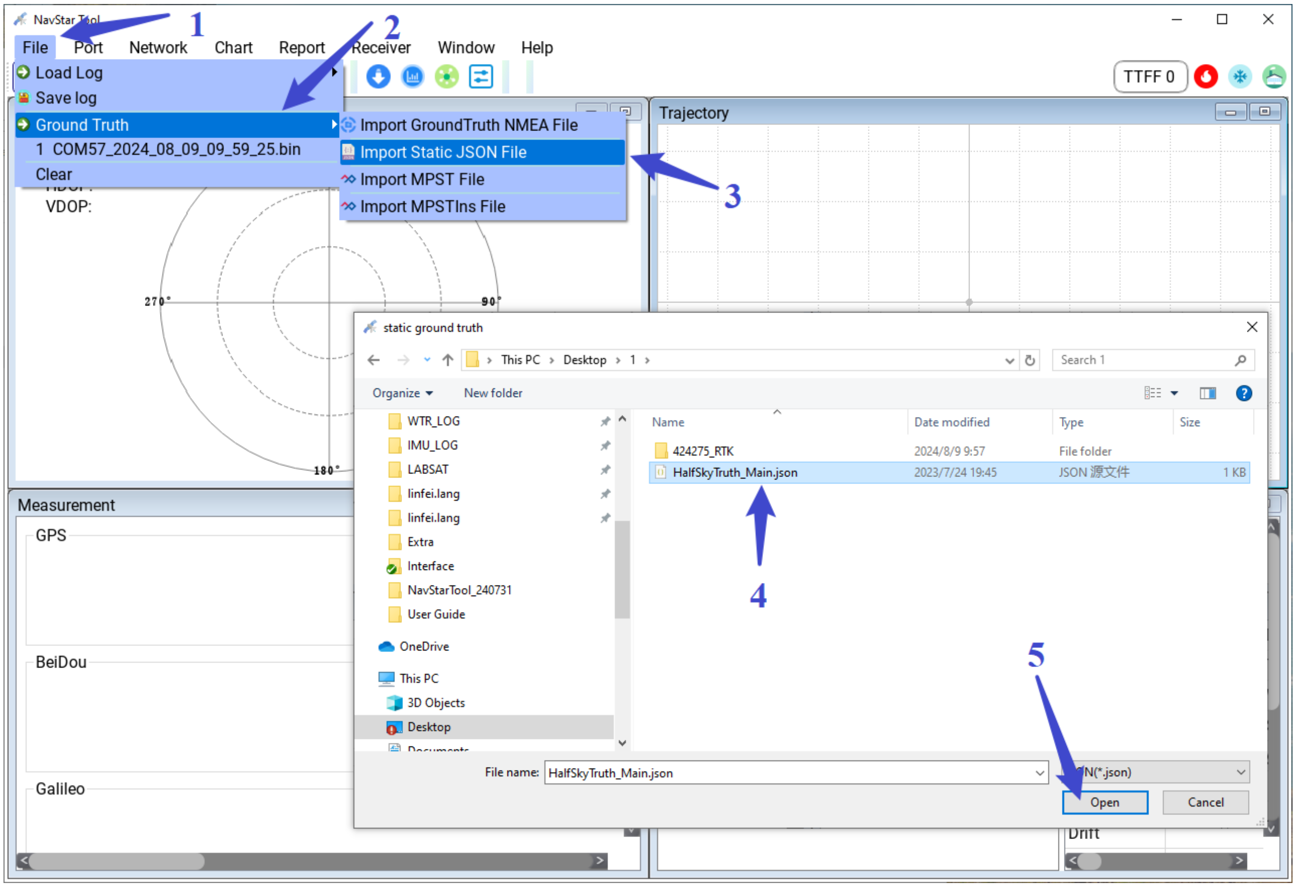
a) NavStarTool Fill in the AGNSS account



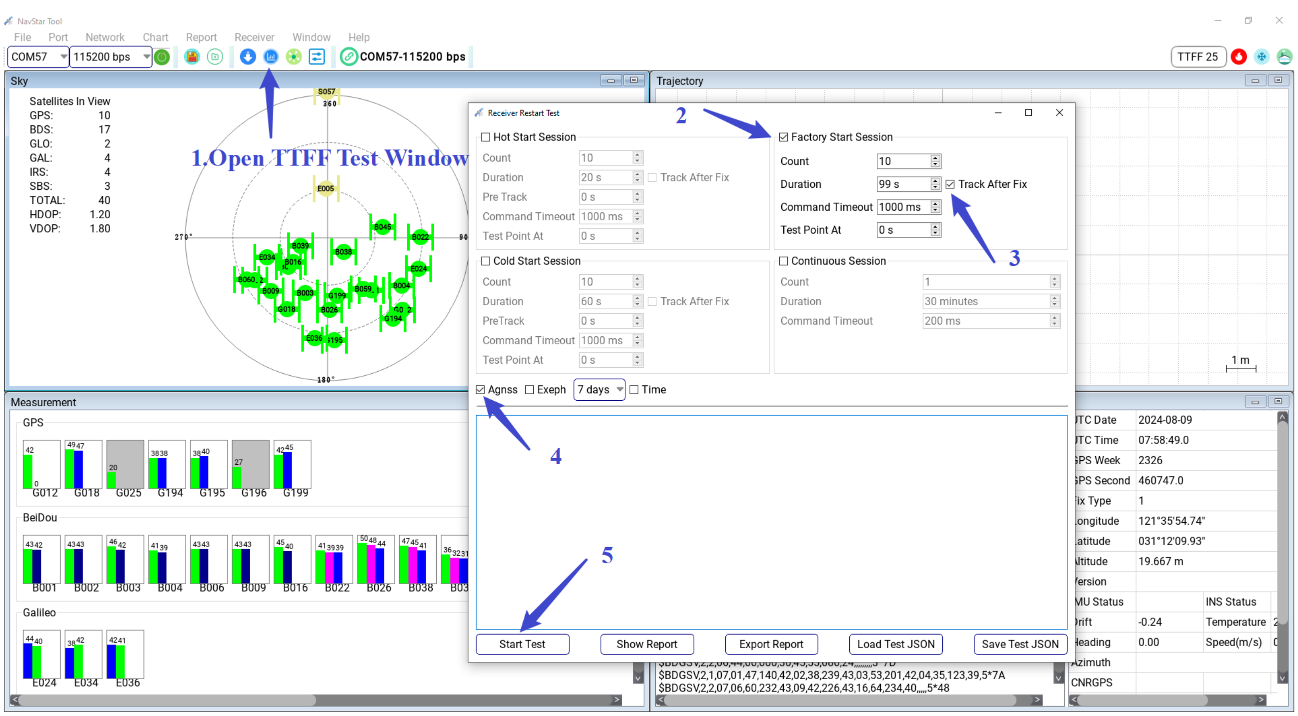
b) Start NavStarTool, Connect the serial port (Refer to section 3.1), and save Log;



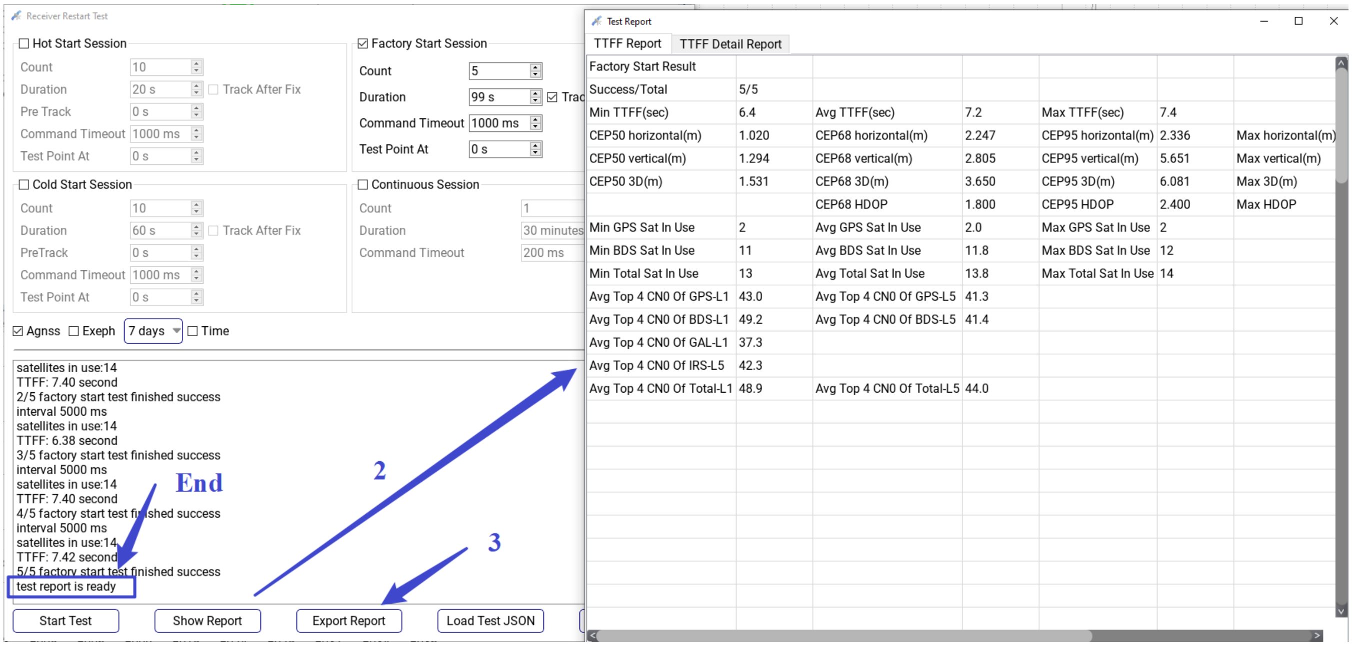
c) Import the truth file, the format see section 8.1 The JSON File Format of Ground Truth(If the truth file is not imported, the result is in line with the accuracy statistics);



d) Open the TTFF Test window, configure the test parameters, start the test, and wait for the end of the test;



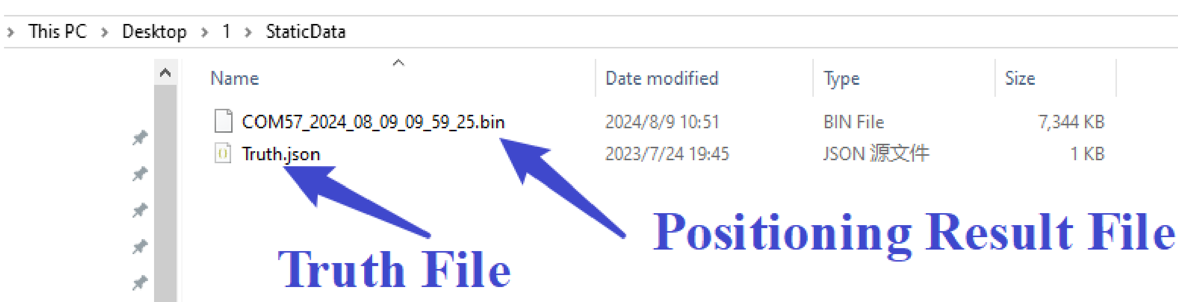
e) Show the results.



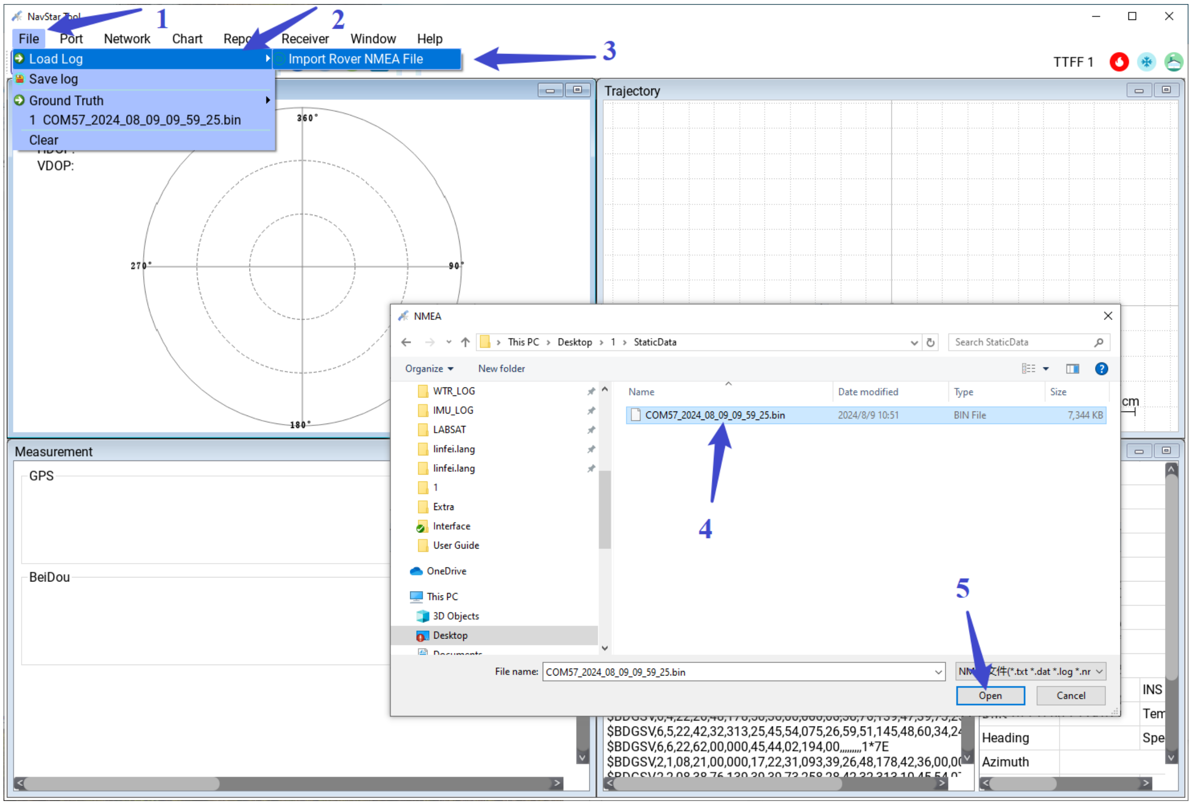
3.9 Positioning Accuracy Analysis

3.9.1 Static Positioning Accuracy Report

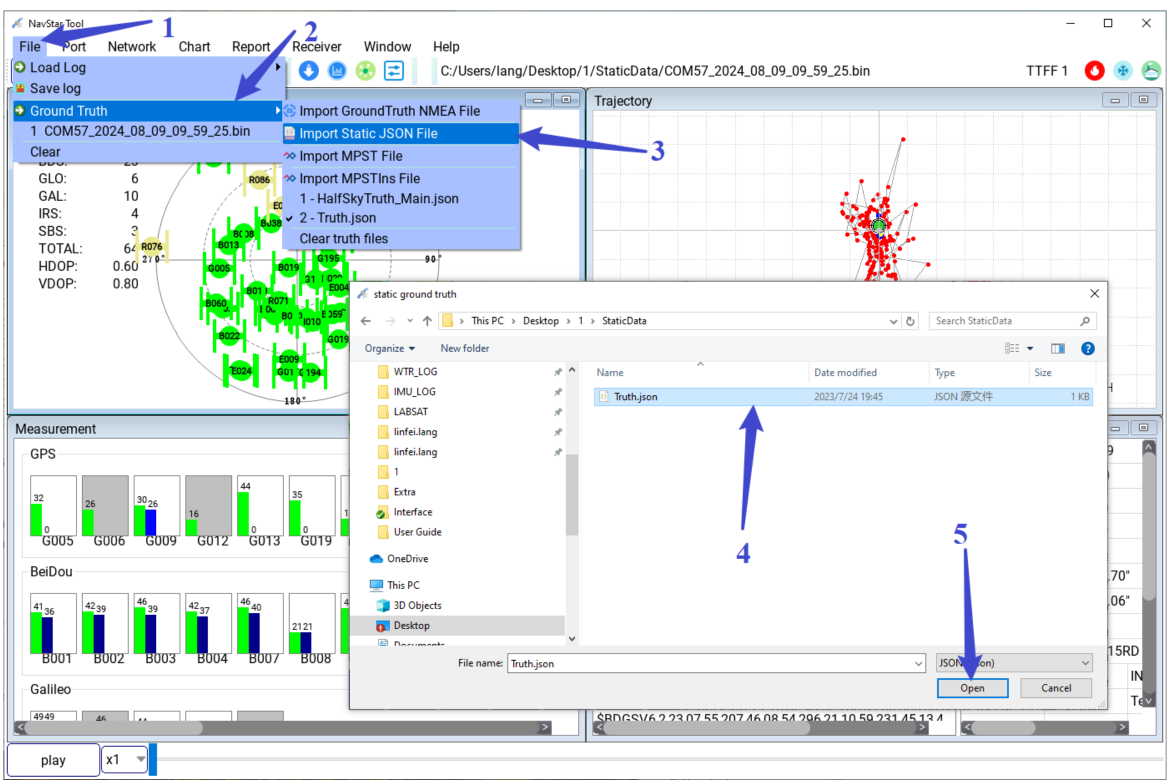
a) Before analyzing positioning accuracy, two files need to be prepared, one is NMEA positioning result file and the other is JSON Truth file. The format is shown in Section 8.1. The JSON format example of Ground Truth file is as follows:



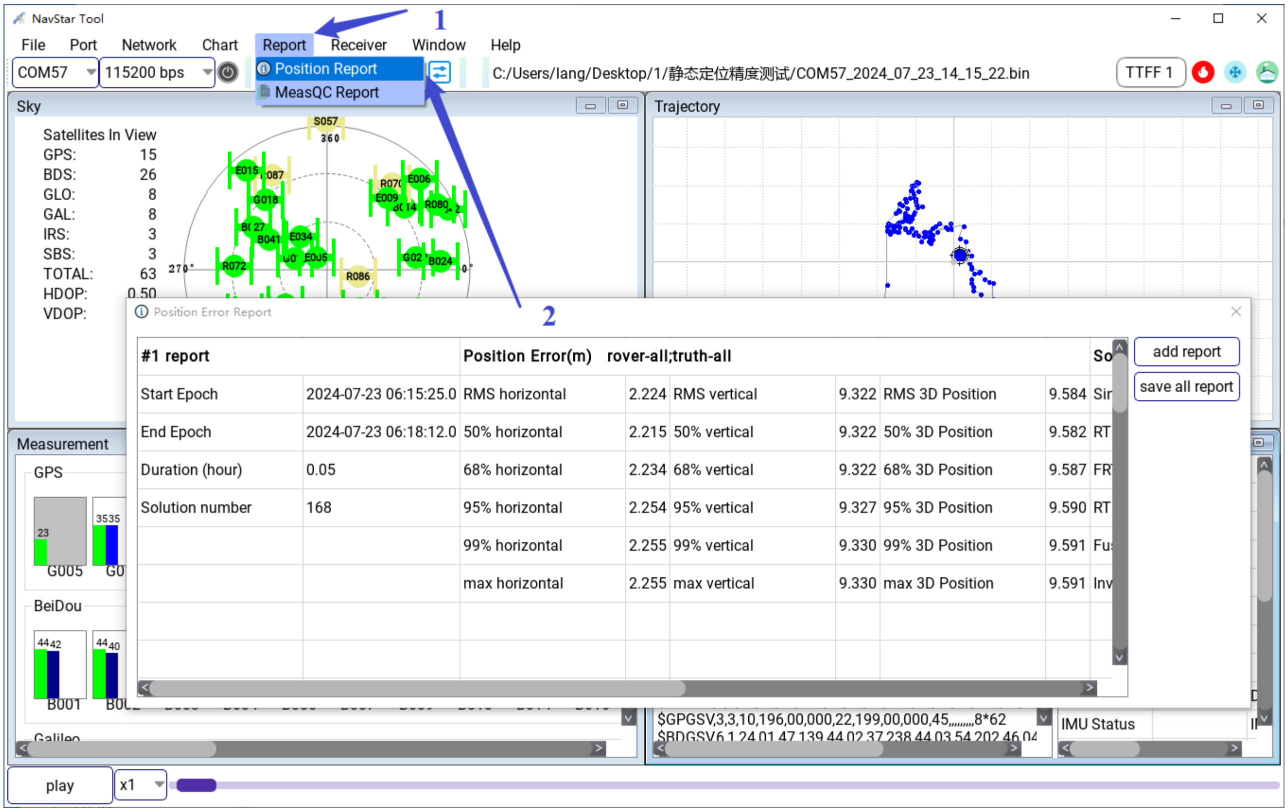
b) Import the positioning result file;



c) Import truth file;

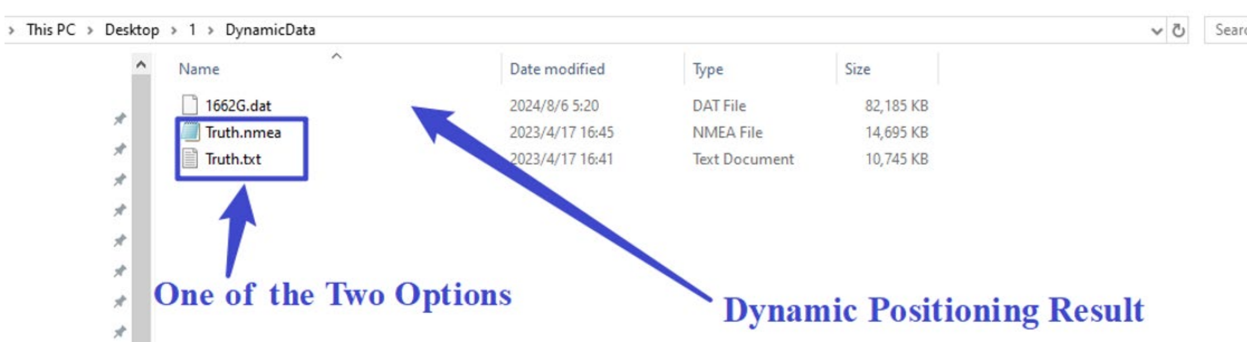


d) Generate reports;

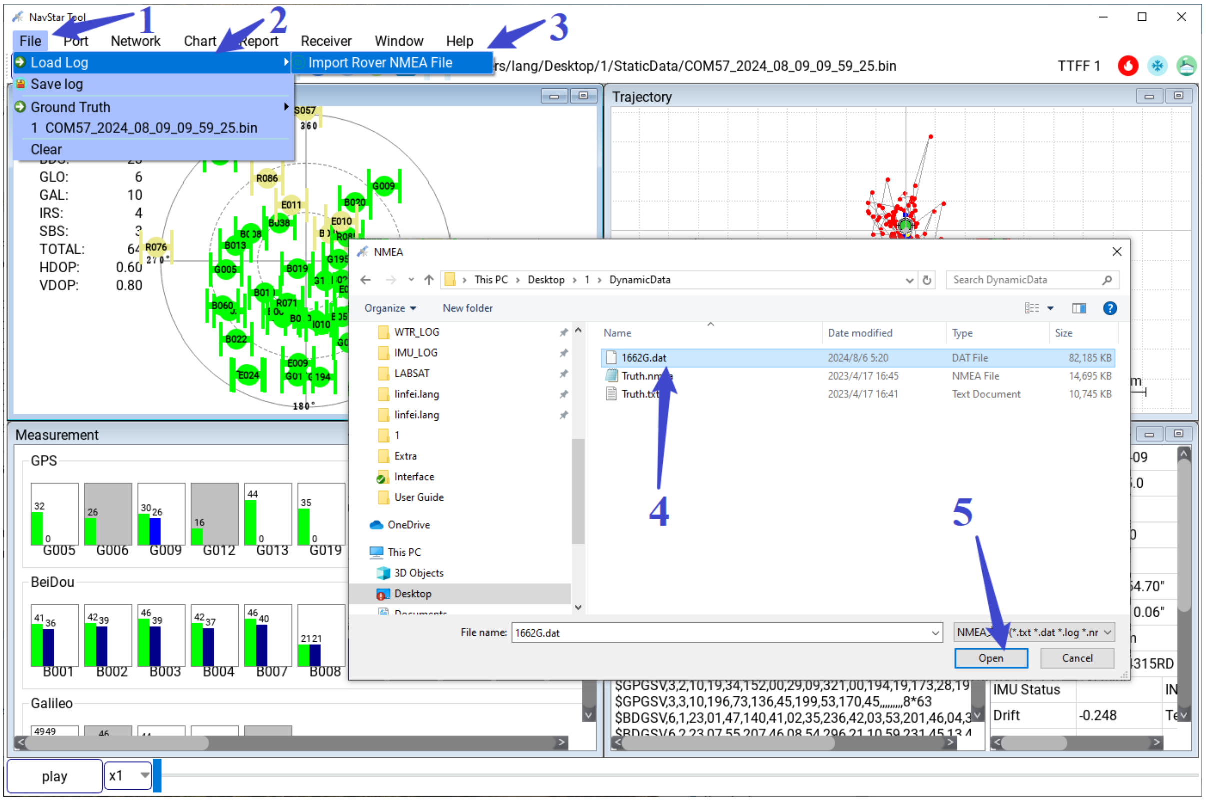


3.9.2 Dynamic Positioning Accuracy Report

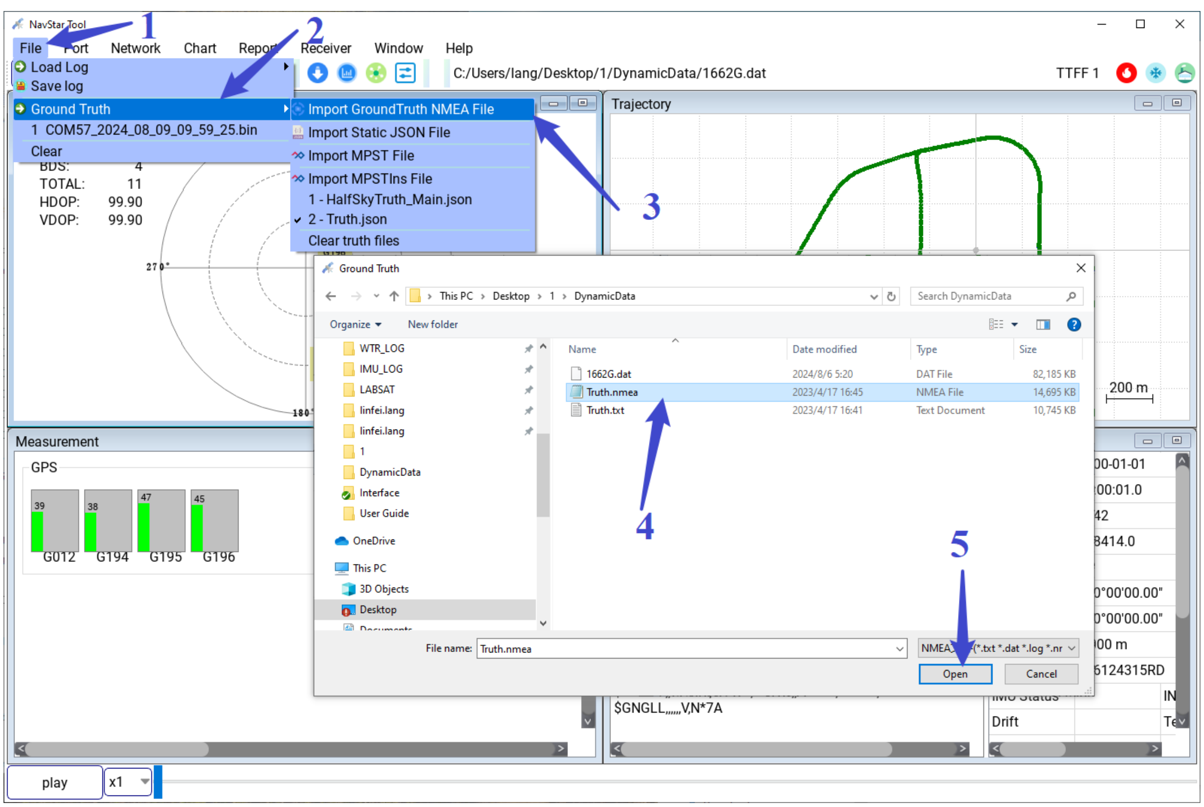
a) Before analyzing positioning accuracy, two files need to be prepared, one is NMEA positioning result file and the other is Truth file, the format can be NMEA format or MPST format (8.2 The MPST File Format of Ground Truth), as shown below:



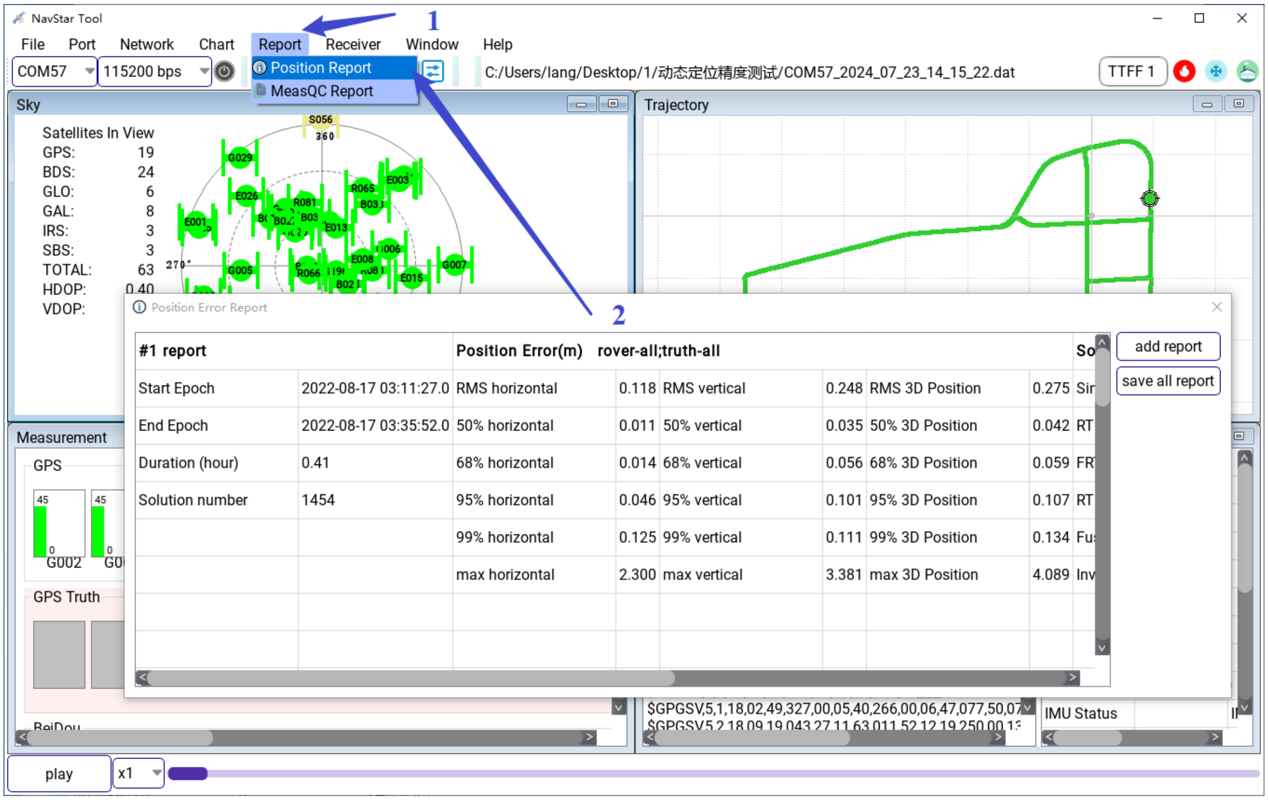
b) Import the positioning result file;



c) Import truth file;



d) Generate reports;

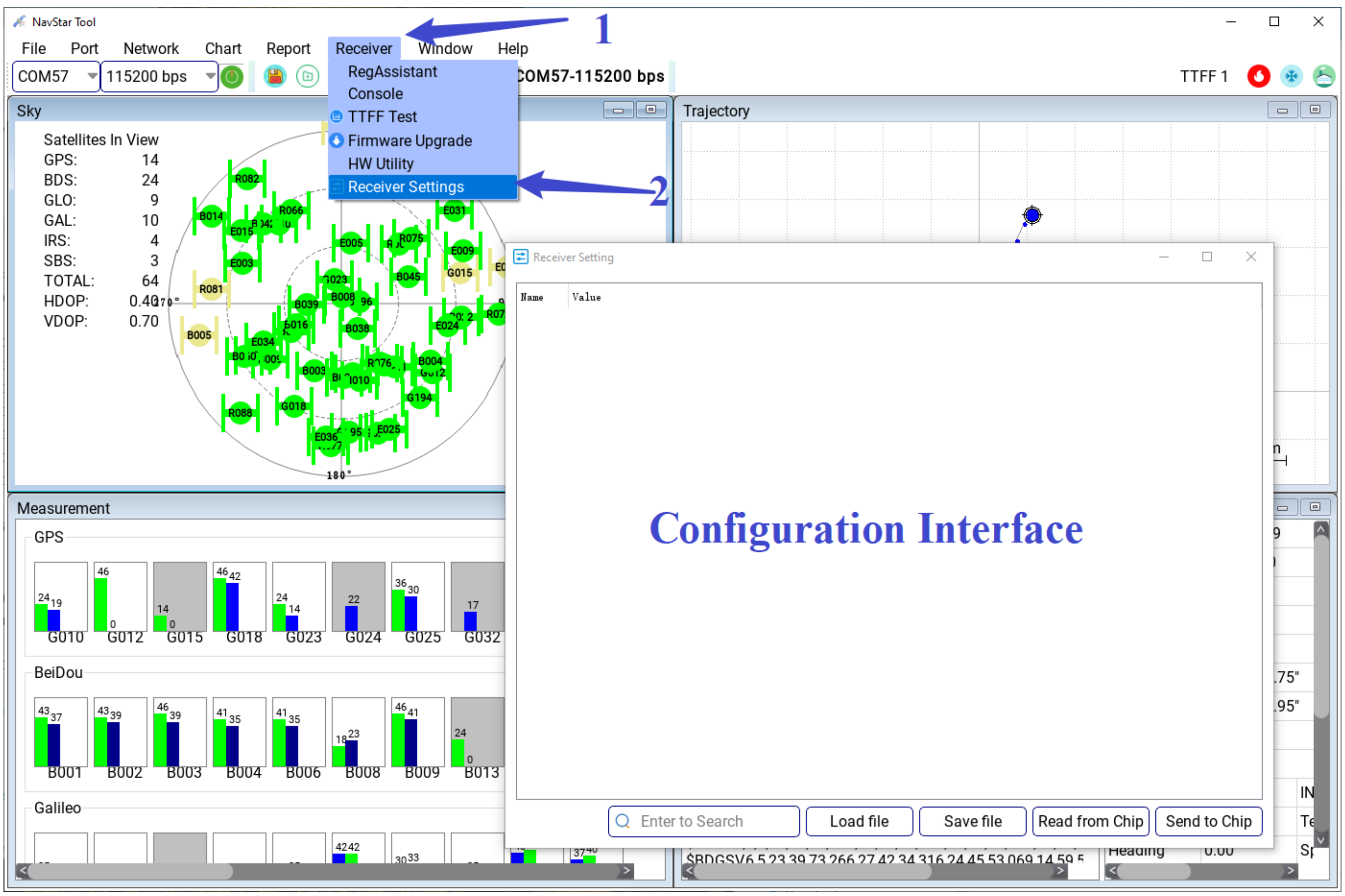


4. Receiver Configuration Tool

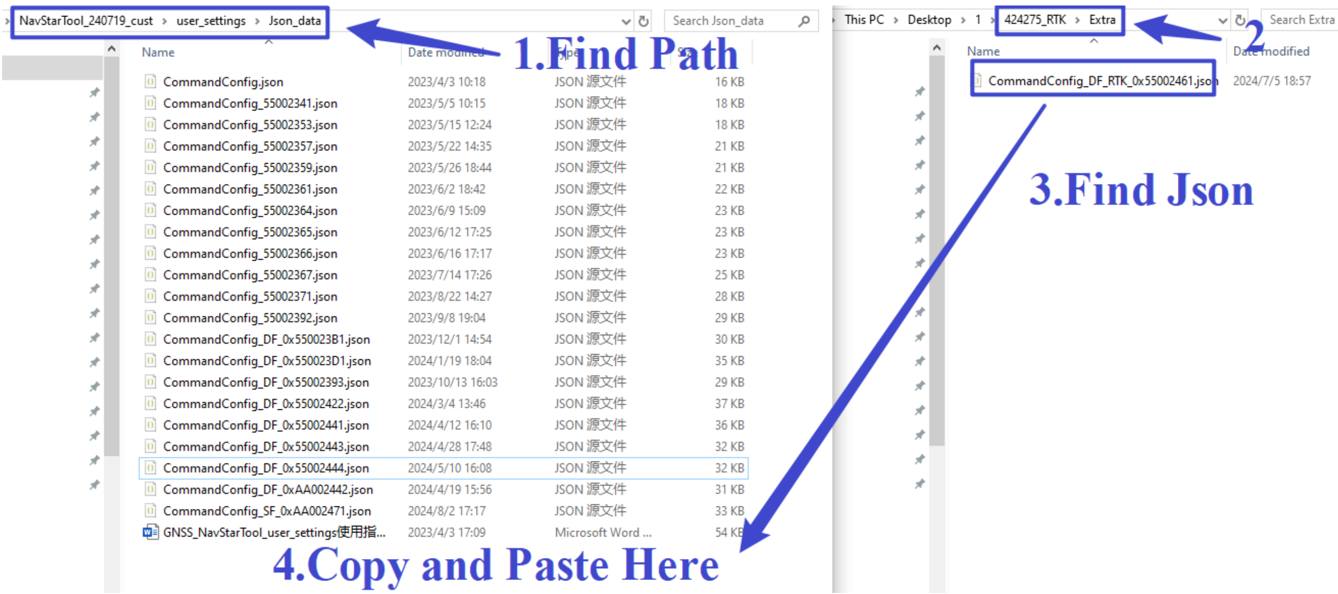
4.1 Configuration Interface

The configuration screen provides a convenient interface for reading and modifying the receiver configuration. It supports multiple configuration parameters, such as working mode, solution mode, and peripheral Settings.

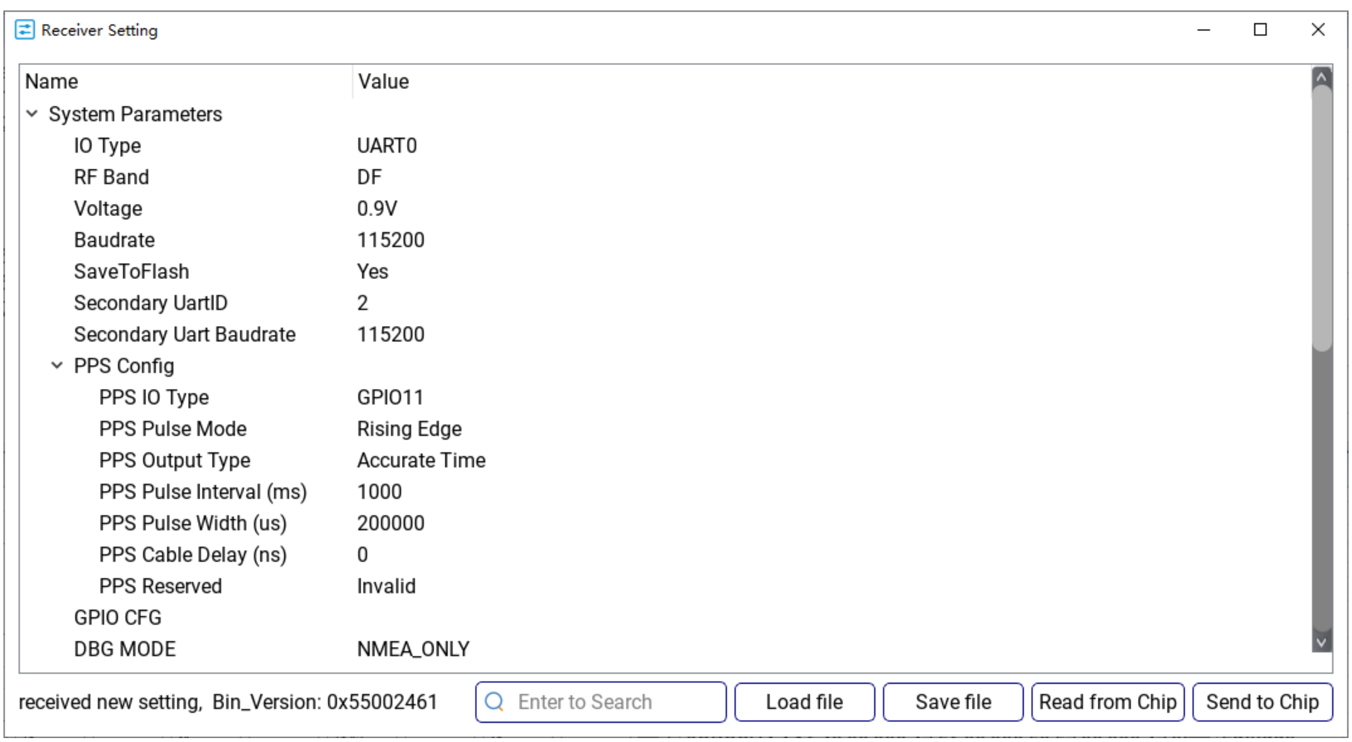
a) Click the menu bar [Receiver] -> [Receiver Settings] or click the shortcut key " " to pop up the receiver configuration interface.



b) No configuration will be displayed after the configuration screen is opened for the first time. Click [Read from Chip] button to read the configuration of the current receiver. If it is successful, "received new setting, Bin_Version:**********" will be displayed; If it fails, "Bin_Version ******** no matched,Please check!" will be displayed on the configuration interface. If the system displays that the surface tool configuration directory does not match the configuration Json module, you need to install the tool "NavStarTool******* \user_settings\Json_data" directory into the provided Json file, as shown below



c) After the configuration file is successfully read, detailed configuration items are displayed. The configuration parameters include multiple types, such as option box, check box, and numeric type. When the mouse hover over the name of the configuration item, the prompt description of the corresponding configuration item will be displayed. Please refer to the prompt content before modifying the configuration.



d) Click the [Send to Chip] button to modify the receiver configuration to the displayed value on the interface. After the configuration is successful, the interface will print "Setting is done successfully ". (All configurations in the interface take effect)

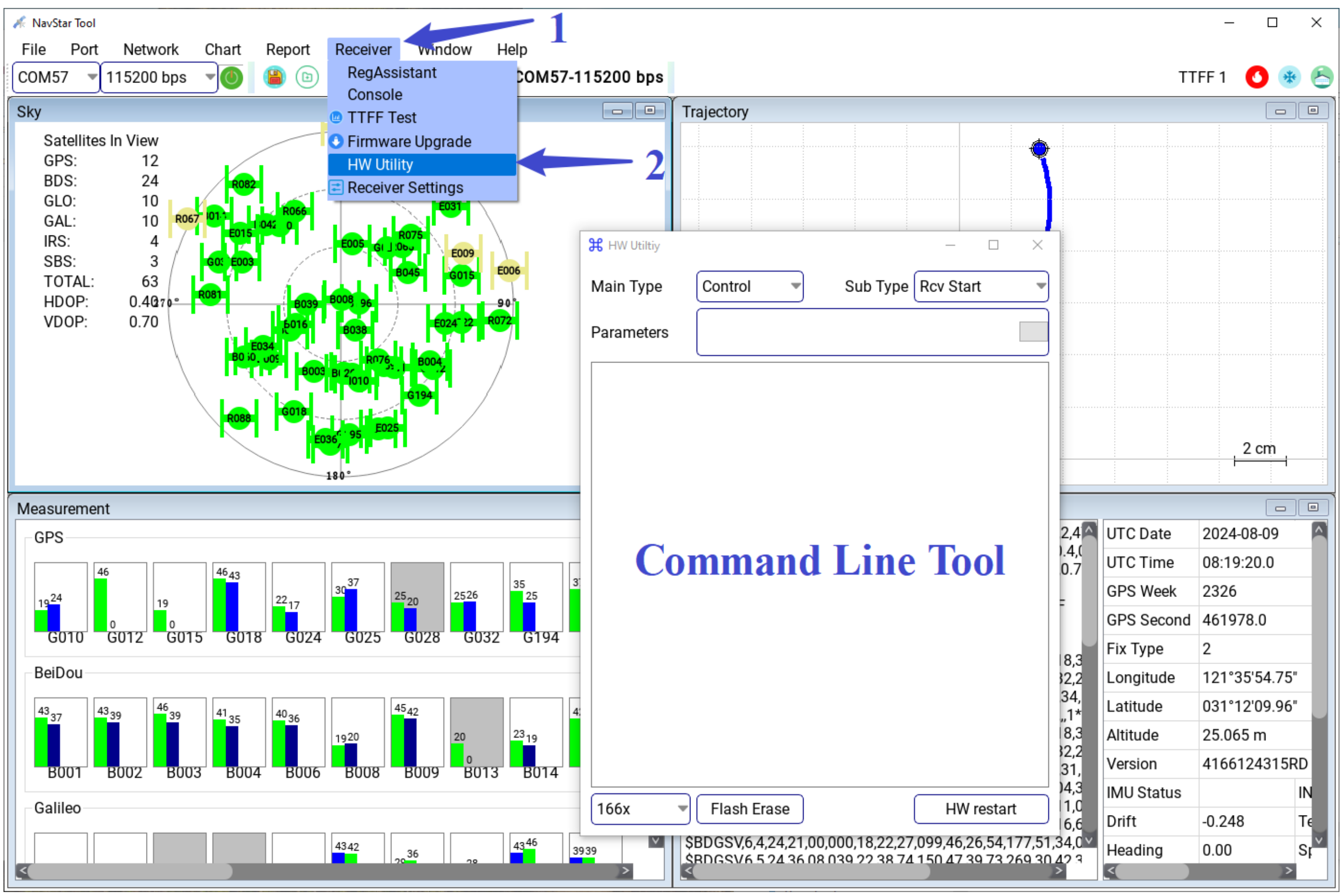
Instructions:

- In principle, it is **not recommended** that users modify the configuration themselves!
- For customers who have customized requirements, the firmware we provide is configured as you need by default.
- [Load file] You can read the json file without connecting the device to check the provided json configuration.
- [Save file] You can adjust the configuration on the interface according to your requirements and save it as a new json configuration file for generating the Full Bin or other purposes

4.2 Command line Tool

The command line tool supports triggering the BK GNSS chip hard restart process, as well as sending other control commands.

Click the menu bar [Receiver] -> [HW Utility], the HW tool window will pop up.



Hard restart process: Click the [HW restart] button in the HW Utility window. The chip will trigger watchdog reset, clear all navigation positioning information, and restart.

Other control commands: After selecting Main Type and Sub Type respectively, click the "" button to initiate the command. If the command contains arguments, do so in Parameters text box: Parameters separated by commas or Spaces, hex data should be preceded by 0x, case insensitive.

Instructions:

Choose between A comma and A space separator, such as "A B" or "A,B". Do not use a comma and a space at the same time.

Control command use cases:

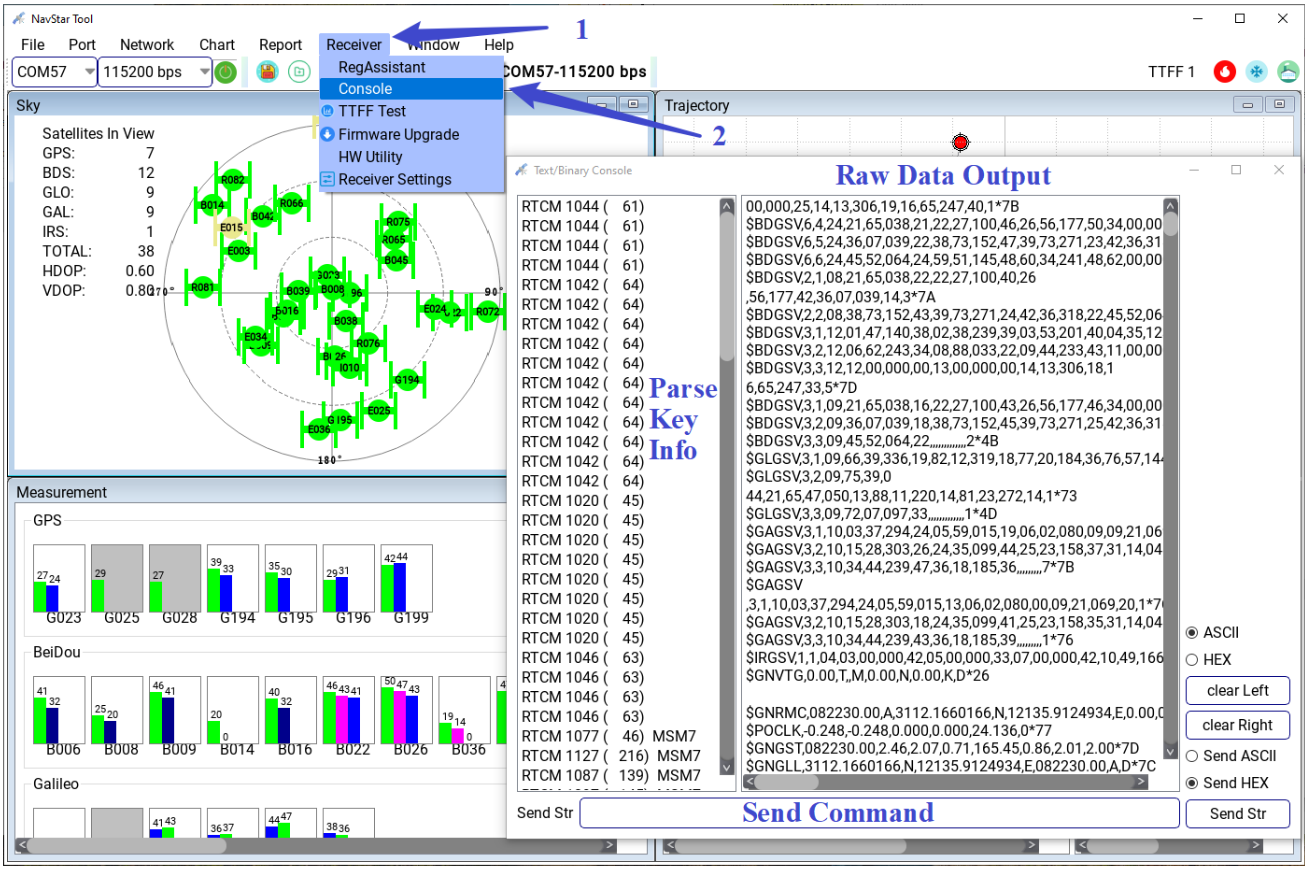
MainType	SubType	Parameters	Description
Control	Rcv Start	-	Start GNSS receiver.
Control	Rcv Stop	-	Stop GNSS receiver.
Control	Set Test Mode	One-Parameter 0~0xFFFF	Set TTFF Test mode: One parameter, 0- hot start test, 0xFF- cold start test, 0xFFFF- factory start test.
Control	WTD Reset	Two-Parameter 1~30 0~2	WatchDog Restart： Two parameters,The former indicates the wait time for the reset (seconds), and the latter indicates whether flash secondary data is erased (0 indicates that Flash secondary data is not erased, 1 indicates that navigation data is erased, and 2 indicates that configuration parameters are erased).
Control	Set Baudrate	One-Parameter 9600/115200/460800/ 1000000/2000000	Set the serial port output baud rate: One parameter, Baud rate (bps). After the setting is successful, the chip restarts. Please reconnect with the newly set baud rate.
Control	Set Output Rate	One-Parameter 1/2/5/10/20	Set the output positioning frequency: One parameter, NMEA output frequency (Hz).
Power	Deep Sleep	-	Set the Deep Sleep mode.
SetTime	Preset XO PPB	Two-Parameter 0~0xFFFFFFFF 0~0xFFFFFFFF	Manually configure the frequency offset PPB (signed) and the preset frequency offset range PPB (unsigned) of the crystal oscillator.

NAV CFG[1]	Save Flash	-	Save all NAV CFG configuration items modified during operation, save them to the Flash, and restart them.
NAV CFG	GNSS Signal	One-Parameter 0x1~0x3FFF	Configure GNSS chip for frequency signal of satellite search positioning.
NAV CFG	Receiver Mode	One-Parameter 0~8	Configure Receiver Mode.
NAV CFG	Position Mode	One-Parameter 0~6	Configure the GNSS chip positioning solution mode.
NAV CFG	Measurement Mode	Three-Parameter 0~2 1/2/5/10/20 0~5	Set whether to set the initial observation quantity and related configuration items.
NAV CFG	Align Time Sys	Two-Parameter 0~5 0~5	Configure the receiver time and PPS time alignment.
NAV CFG	Preset Gps Week	One-Parameter 0~65535	Configure the default GPS Ephemeris Week Number to solve the cross-loop problem.
NAV CFG	Low Level Config	Three-Parameter 0~1 1~2 8~9	Configure the underlying options related to initialization, including the configuration of output mode, radio frequency, and operating voltage.

Note: Please refer to the < BK166X Series Interface Specification.docx > for detailed description of all NAV CFG configuration item parameters.

4.3 Serial debugging Assistant

Click the menu bar [Receiver] -> [Console], the serial debugging assistant tool window will pop up, as shown below.



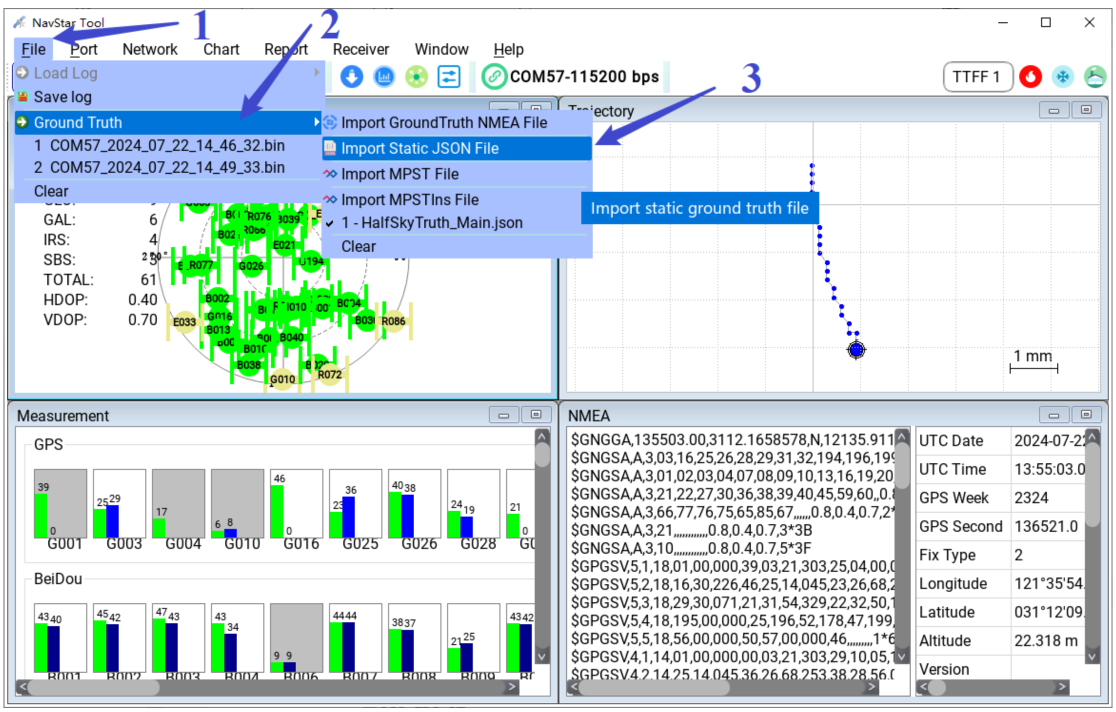
The window can customize the command sending, the default is Hex sending, the check box on the right can choose ASCII mode sending or display;

5. Positioning Test Tool

Positioning Test Tool supports start positioning (TTFF) test and Continuous tracking positioning (Continuous) test, and generates precision statistics report after the test.

5.1 Import Truth File

Click the menu bar [File] -> [Ground Truth], select [Import Static JSON File] for static scenes, select [Import GroundTruth NMEA File] or [Import MPST File] for dynamic scenes; If no truth file is imported, the result conforms to the accuracy in statistics;

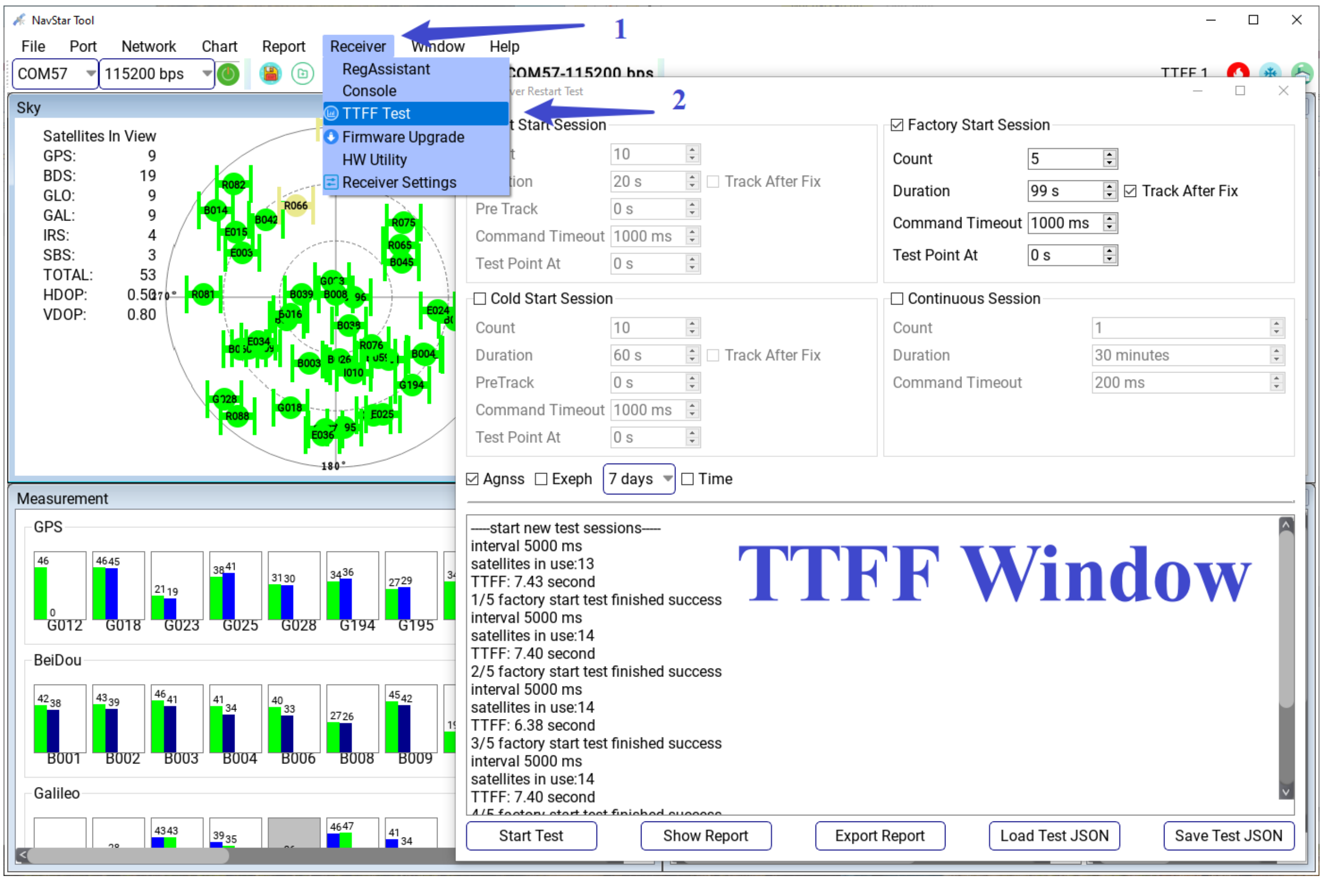


Instructions:

- Static reference coordinates JSON file format please refer to Section 8.1 The JSON File Format of Ground Truth;
- For dynamic testing, the reference Truth supports NMEA format and MPST high-precision integrated navigation format, the format refer to Section 8.2 The MPST File Format of Ground Truth;
- If there is no truth file, the tool will use the average value of GNSS chip positioning results as a reference to calculate the coincidence accuracy errors in positioning.

5.2 Parameter Configuration

Click the menu bar [Receiver] -> [TTFF Test] or shortcut key “ ” to go to the page of Receiver Restart Test.



The “Receiver Restart Test” window is divided into two parts. The upper part is the control area, which is used for setting parameters of TTFF test. The lower part is the display area, which displays the current positioning phase information in real time. After the test is completed, you can click the [Show Report] button below the display area to display the positioning result statistics Report, and the [Export Report] button to export the report to excel file.

The control area is divided into four test modes: hot start positioning, cold start positioning, factory start positioning and continuous tracking positioning, each with its own configuration parameters. "Count" indicates the number of sessions required for the current mode. Here we define a session to initiate a positioning, that is, a GNSS receiver from a start to stop working process. To ensure the validity of statistical results, it is recommended to set the number of sessions to no less than 30.

“Duration” indicates the duration of each session. If "Track After Fix" is checked, the receiver continues to work after the current session is successfully located until the time set by Duration is satisfied. Otherwise, the receiver will stop immediately after the successful positioning. To ensure the consistency of the test, you are advised to check it.

"Interval" indicates the time between two adjacent sessions, that is, the time between when the previous receiver stops working and when the next one starts working. After the check box "Enable Random" is checked, each interval is a random number between 0 and X; Otherwise, the constant X. Where X is the time set by "Interval". To ensure the randomness of the test, you are advised to select.

“Pre-track” indicates the pre-tracked time before the current test is performed only once, and the data during this time will not be included in the positioning result statistics. This parameter is mainly used to collect ephemeris book data in hot start mode, and it is recommended that the hot start be set to 300 seconds.

"Command Timeout" indicates the waiting timeout period for the tool to initiate each command. The default is 1 s.

“Test Point At” indicates the epoch time of locating statistical indicators in each session. If positioning is successful before this time, only the positioning result at the set time is selected for precision statistics. If it has not been positioned at this time, the precision statistics will select the positioning result of the first positioning moment. If you do not check, the precision of the first positioning point is collected. This option does not affect TTFF statistics.

Click [Start] button to start the test, click [Show Report] button to view the statistics Report, and click [Export Report] button to export the report.

5.3 Test Case

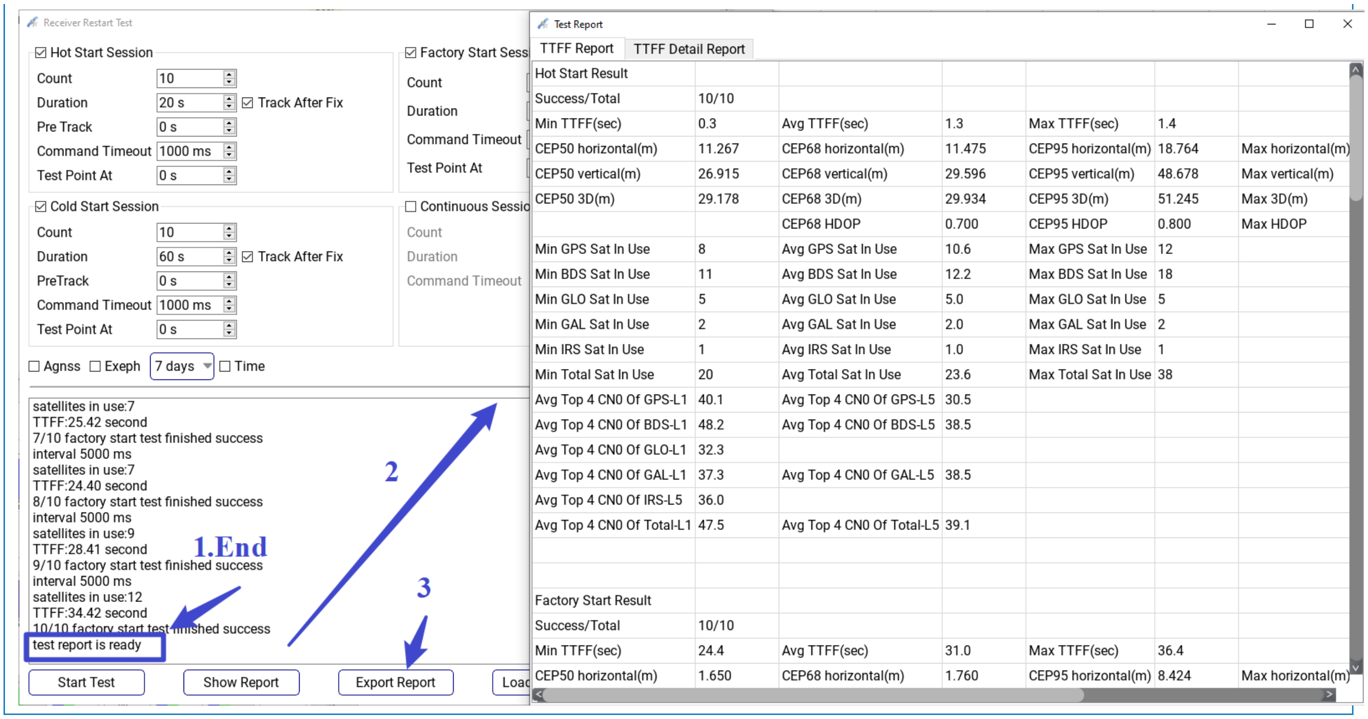
- **Factory startup positioning test cases**

a) Connect the serial port;

b) Import the reference coordinate truth file;

c) TTFF test select Hot/Cold/Factory Start Session:For open-sky strong signal scenarios, the default parameters can be used; For other scenarios, it is recommended that Duration parameters be set to 60 seconds for Hot, 120 seconds for Cold, and 180 seconds for Factory. For other parameters, refer to the preceding section

d) Click [Start Test] to start the test, wait until the test is over to view and export the report.



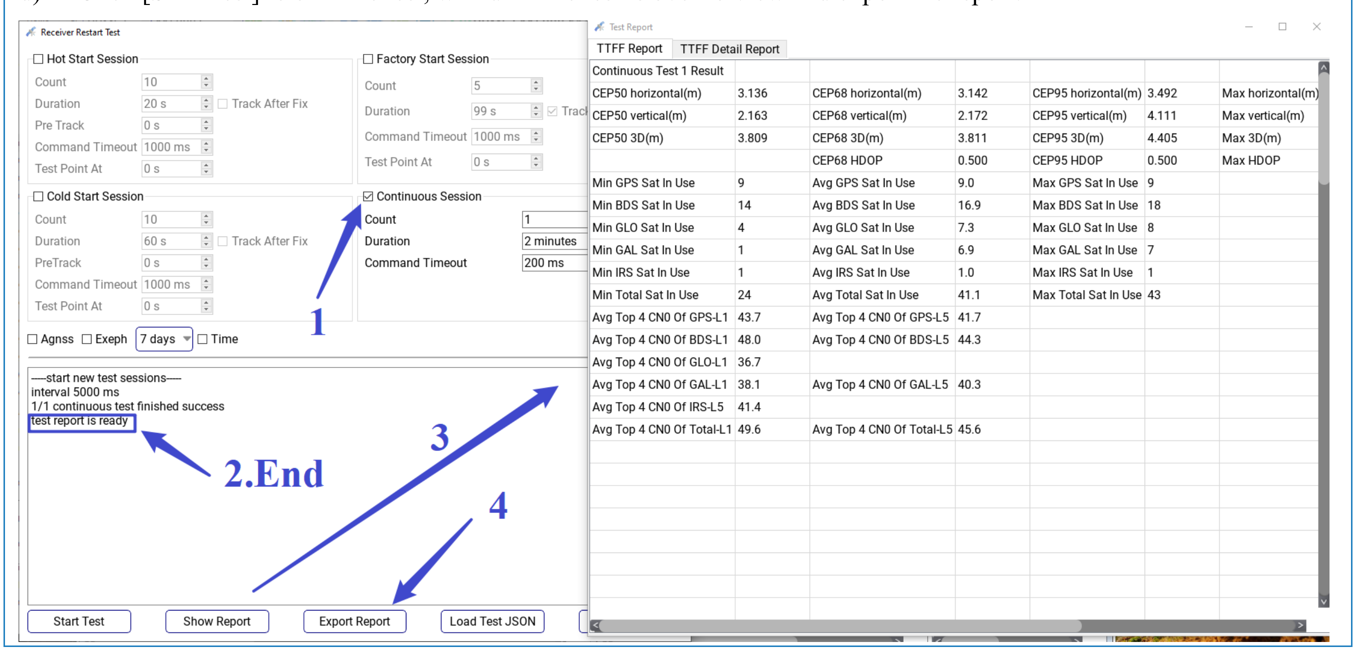
- **Continuous tracking positioning test cases:**

a) Connect the serial port;

b) Import the reference coordinate truth file;

c) TTFF test select Continues： Duration is recommended to be at least 1 minute. For other parameters, please refer to the above description;

d) Click [Start Test] to start the test, wait until the test is over to view and export the report.

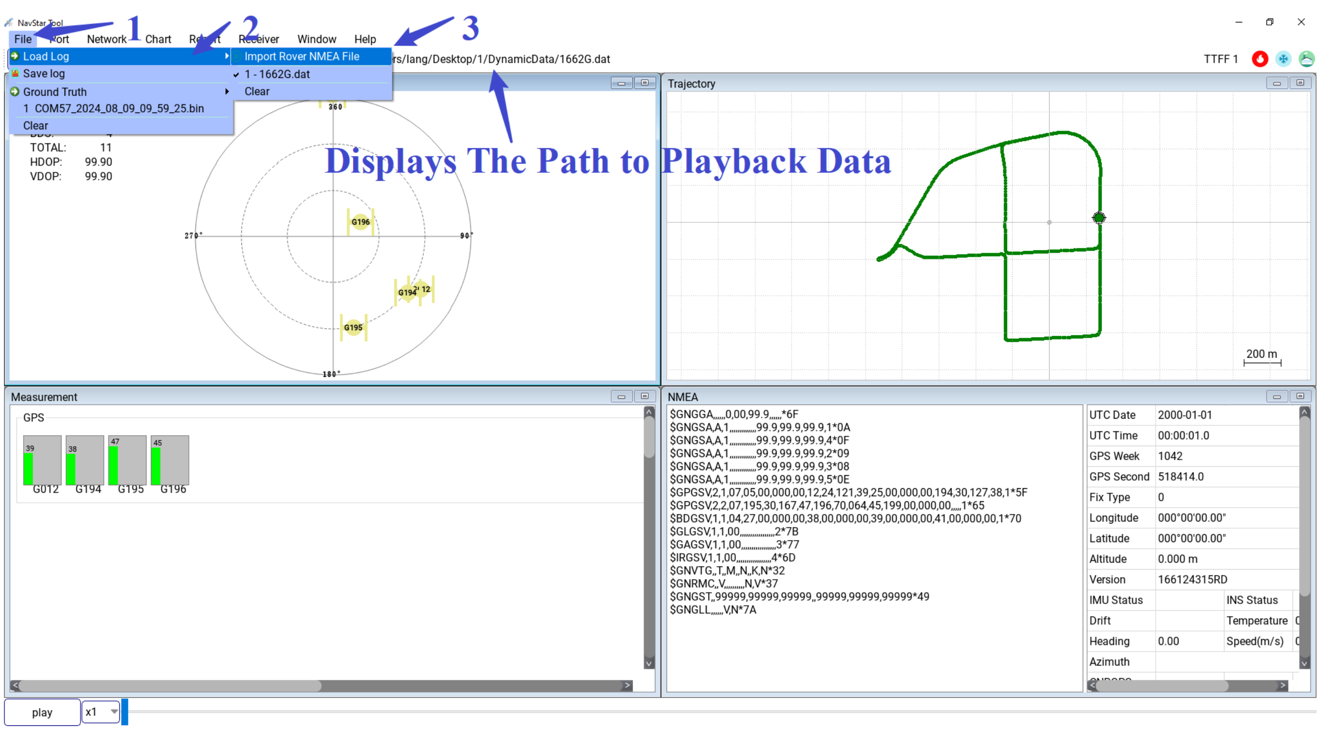


6. Play-Back and Accuracy Statistics

The GNSS NavStarTool supports offline mode. After importing historical NMEA files, the GNSS NavStarTool can play back NMEA log data. At the same time, it also supports the output of the positioning error distribution graph of the historical NMEA file and the accuracy statistics report (Need to import truth file).

6.1 Import NMEA Log

a) Click on the menu bar [File] -> [Load Log] -> [Import Rover NMEA File] and select the NMEA file to be played back.



b) After reading the file, the main interface will display GNSS output information, please refer to Chapter 2 for details;

c) Meanwhile, the playback navigation bar at the bottom of the main window is enabled. Drag the slider to display the positioning output information of the playback file at any time, refer to Section 3.6 Data Playback.

Instructions:

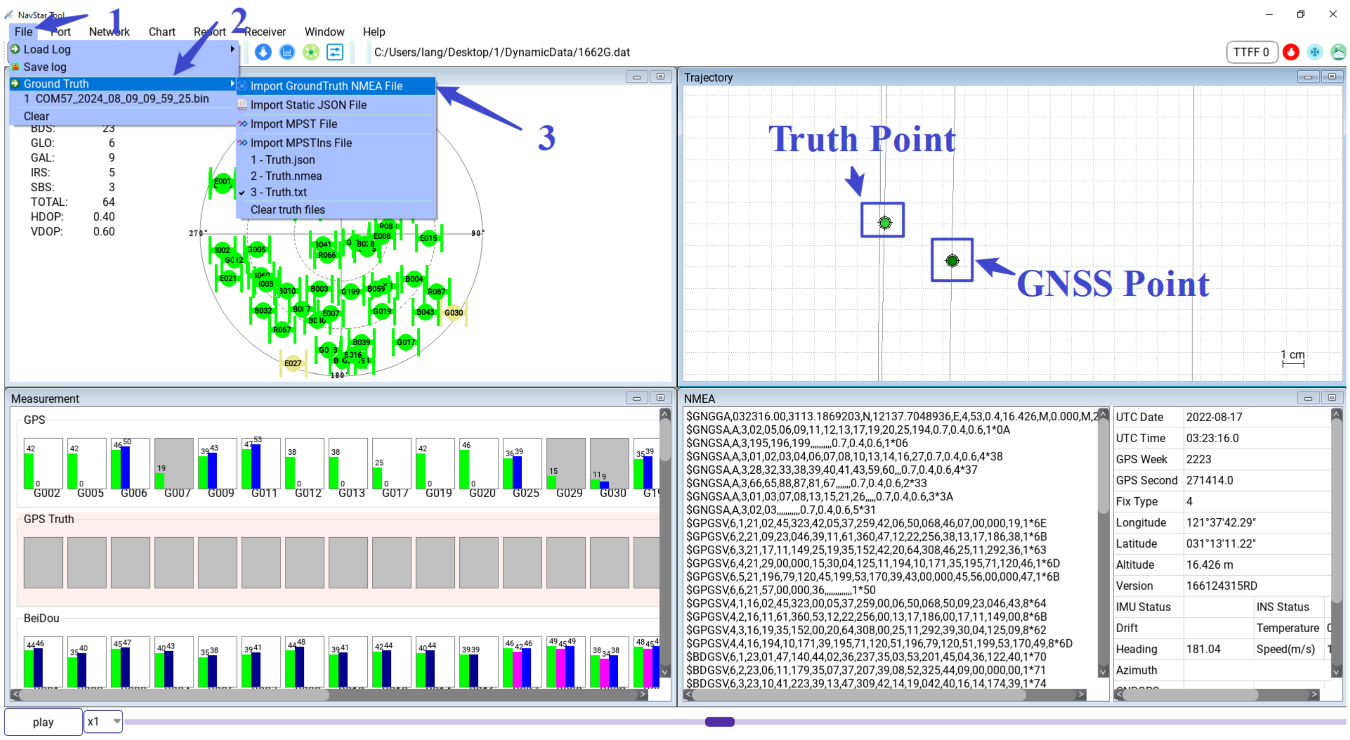
- The playback file must contain GGA and RMC statements, otherwise there will be an exception to the accuracy statistics.
- It is recommended that the playback file contain GSV and GSA statements, otherwise the star map display will be abnormal.
- The "$" sign must be at the beginning of the line in the return file, otherwise it will not be parsed.

6.2 Import the Reference Truth File

The method of importing truth file is the same as that in 5.1 Import Truth File.

Click the menu bar [File] -> [Ground Truth], For Static scene， select [Import Static JSON File], For dynamic scenes, select [Import GroundTruth NMEA File] or [Import MPST File] or [Import MPSTIns File] . See the Appendix of 8.1~8.3 for the format description. The main screen track display screen will synchronously.

Display the track of the truth file, the track color is slightly lighter than the playback file.

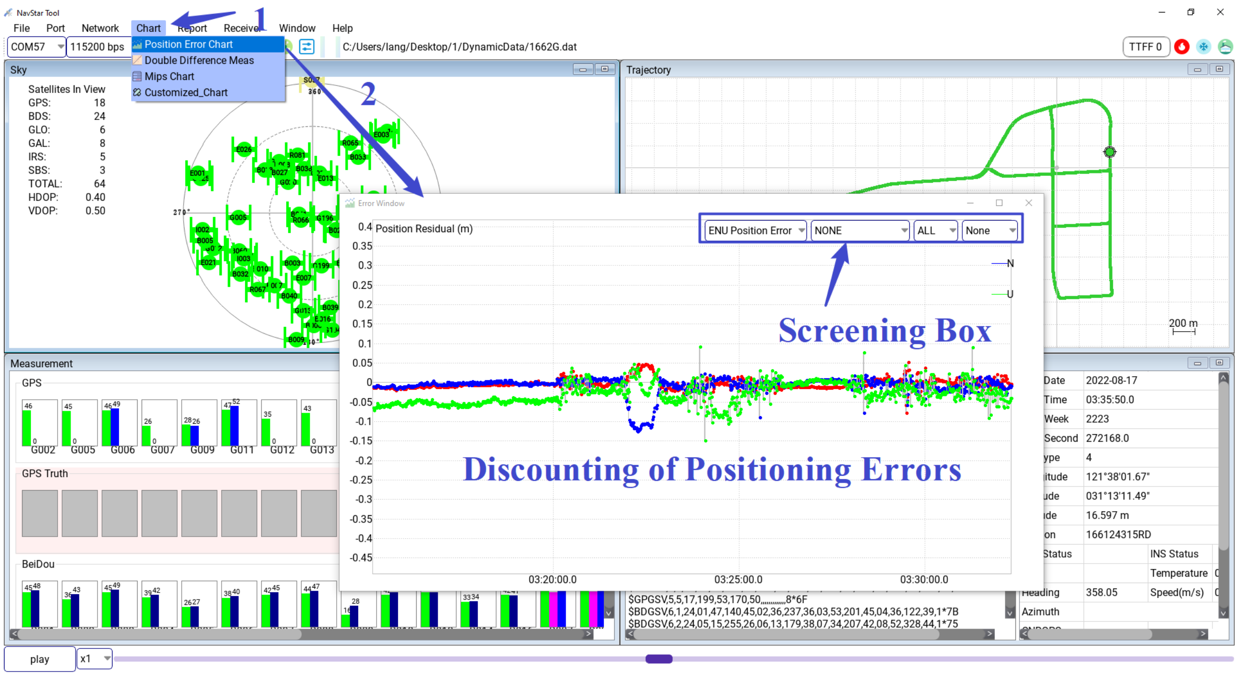


Instructions:

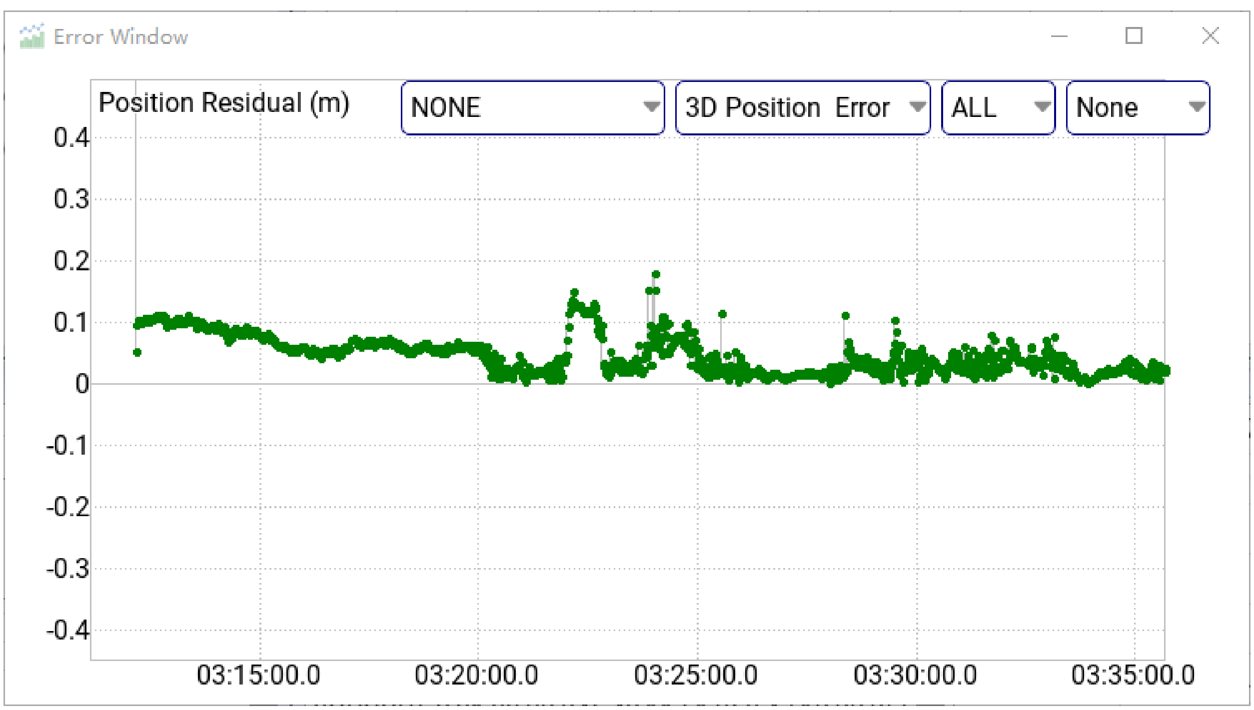
If no truth file is imported, precision statistics will output invalid values.

6.3 Positioning Error Chart

Click the menu bar [Chart] -> [Position Error Chart], the positioning error chart window will pop up;

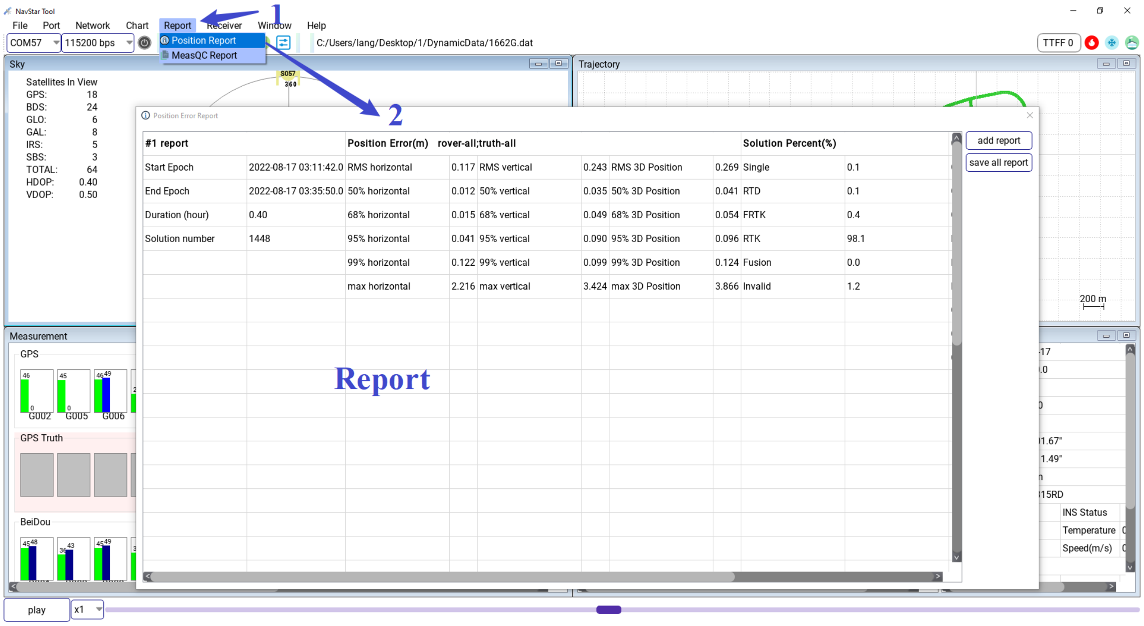


Positioning error chart, horizontal axis is UTC time, vertical axis is positioning error (m). The error curve in the east/north/vertical (E/N/U) direction is displayed by default. The upper right selection box optionally displays horizontal and 3D error curves.

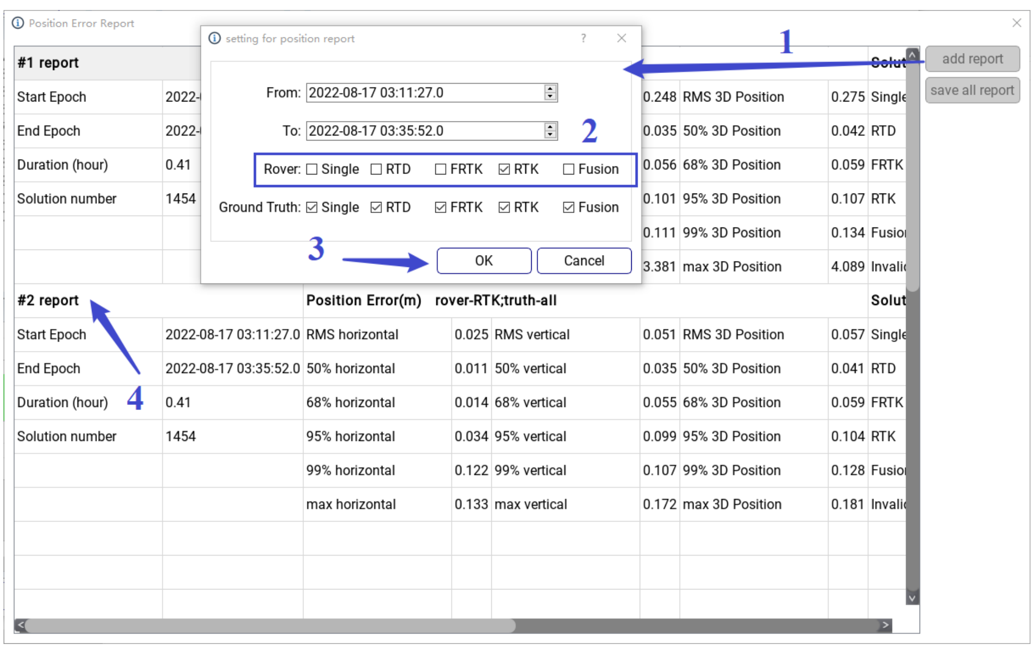


6.4 Positioning Error Accuracy Report

a) Click the menu bar [Report] -> [Position Report], the error accuracy report window will pop up;



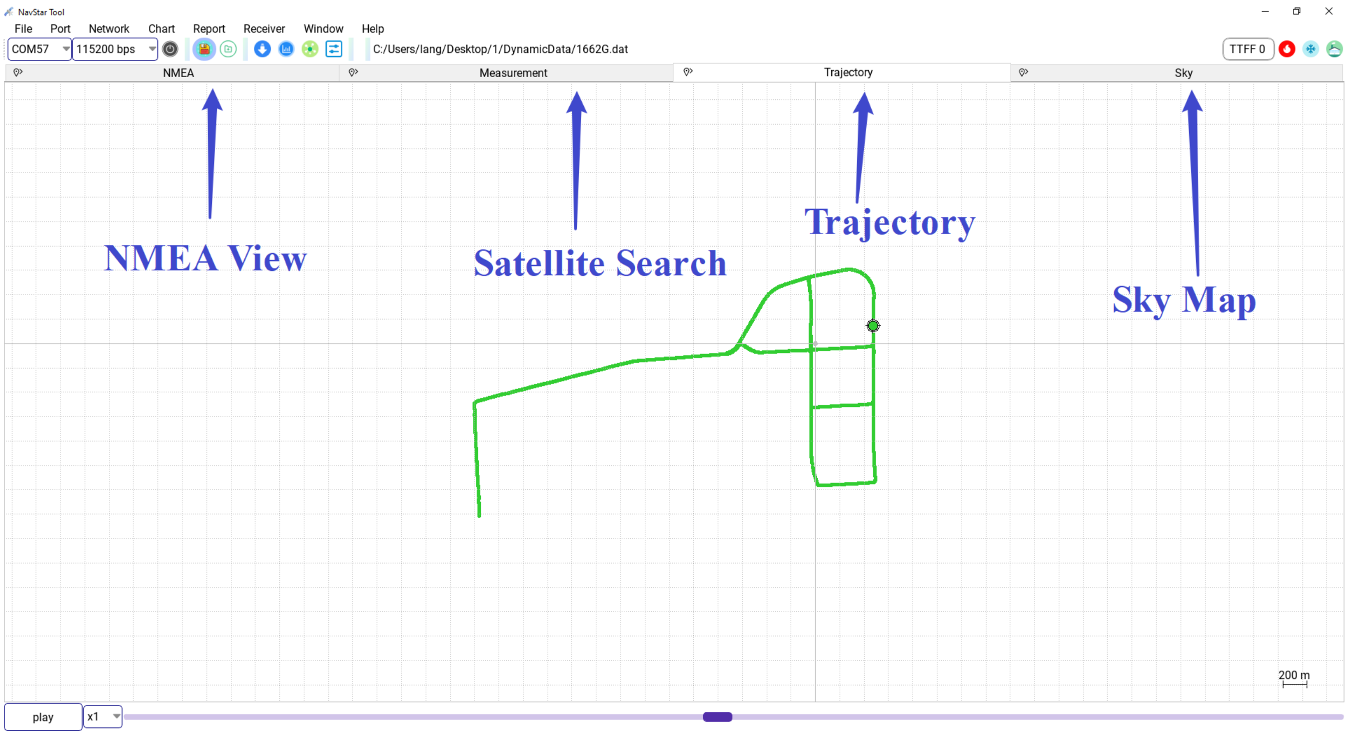
b) The type of positioning solution, horizontal positioning accuracy index (CEP68/CEP95/CEP99/ maximum), vertical positioning accuracy index and 3D positioning accuracy index are calculated in the report; If you need to filter the type of solution, click [add report], you can screen the time and state according to your own needs, and generate a report, as shown below:



c) Click the [Save Report] button to export the report to excel file.

7. Window Arrangement

GNSS NavStarTool supports two different window arrangement modes, Namely Tilt Layout and Tab Layout.Click the menu bar [Window] and select the window Layout mode. The picture below is Tab Layout view.



8. Appendix

8.1 The JSON File Format of Ground Truth

Create a new file ” truth.json”, with the following contents (pos1 latitude, pos2 longitude, unit is degrees; pos3 is elevation, unit is meters) :

```
{
    "type":"LLA",
    "pos1":31.211653265,
    "pos2":121.593542691666667,
    "pos3":36.9441
}
```

Note: The symbols in the file are English symbols, and the format strictly refers to the example; JSON file can be created using notepad to create text, modify the text suffix "*.json".

8.2 The MPST File Format of Ground Truth

```
-----------------------------------END OF HEADER--------------------------------------

 Week    GPSTime          ECEF X          ECEF Y          ECEF Z      Vel-E      Vel-N    Vel-U   posQuality HorizontalAccuracy ElevationAccuracy
 (Wk)      (sec)             (m)             (m)             (m)      (m/s)      (m/s)    (m/s)      (value)        (m)      (m)
 2321 355733.000    -2833696.805     4673055.232     3277365.884      0.000      0.000   -0.001            1    0.018    0.016
 2321 355733.200    -2833696.805     4673055.232     3277365.884      0.000      0.000   -0.001            1    0.017    0.016
 2321 355733.400    -2833696.805     4673055.232     3277365.884      0.000      0.000   -0.001            1    0.017    0.016
 2321 355733.600    -2833696.805     4673055.231     3277365.884      0.000      0.000   -0.001            1    0.017    0.016
 2321 355733.800    -2833696.805     4673055.231     3277365.884      0.000      0.000   -0.000            1    0.016    0.016
 2321 355734.000    -2833696.805     4673055.231     3277365.884      0.000      0.000   -0.000            1    0.016    0.015
 2321 355734.200    -2833696.805     4673055.231     3277365.884     -0.000      0.000   -0.000            1    0.016    0.015
 2321 355734.400    -2833696.805     4673055.231     3277365.883     -0.000      0.000   -0.000            1    0.016    0.015
 2321 355734.600    -2833696.805     4673055.231     3277365.883     -0.000      0.000   -0.000            1    0.016    0.015
 2321 355734.800    -2833696.805     4673055.231     3277365.883     -0.000      0.000   -0.000            1    0.015    0.015
 2321 355735.000    -2833696.805     4673055.231     3277365.883      0.000      0.000   -0.000            1    0.015    0.015
 2321 355735.200    -2833696.805     4673055.231     3277365.883     -0.000      0.000   -0.000            1    0.015    0.015
 2321 355735.400    -2833696.805     4673055.231     3277365.883      0.000      0.000   -0.000            1    0.015    0.014
 2321 355735.600    -2833696.805     4673055.231     3277365.883      0.000      0.000   -0.000            1    0.015    0.014
 2321 355735.800    -2833696.805     4673055.231     3277365.883      0.000      0.000   -0.000            1    0.015    0.014
 2321 355736.000    -2833696.805     4673055.231     3277365.883      0.000      0.000   -0.000            1    0.014    0.014
 2321 355736.200    -2833696.805     4673055.231     3277365.883     -0.000      0.000   -0.000            1    0.014    0.014
 2321 355736.400    -2833696.805     4673055.231     3277365.883      0.000      0.000   -0.000            1    0.014    0.014
 2321 355736.600    -2833696.805     4673055.231     3277365.883      0.000      0.000   -0.000            1    0.014    0.014
 2321 355736.800    -2833696.805     4673055.231     3277365.883      0.000      0.000   -0.000            1    0.014    0.014
 2321 355737.000    -2833696.805     4673055.231     3277365.883      0.000      0.000    0.000            1    0.014    0.014
 2321 355737.200    -2833696.805     4673055.231     3277365.883      0.000      0.000    0.000            1    0.014    0.014
 2321 355737.400    -2833696.805     4673055.231     3277365.883      0.000      0.000    0.000            1    0.014    0.014
```

Note: The file name extension can be "*.txt".

8.3 The MPSTIns File Format of Ground Truth

```
---------------------------------END OF HEADER-------------------------------------

Week    GPSTime        Latitude        Longitude    H-Ell    Vel-N    Vel-E    Vel-D        Roll       Pitch      Heading    Q  posQuality  SatNum  SatUse
(Wk)      (sec)           (deg)            (deg)      (m)    (m/s)    (m/s)    (m/s)       (deg)       (deg)        (deg)          (value) (value) (value)
2321 355733.000   31.1207227272   121.2322291703   13.167    0.000    0.000    0.001  1.12300908  3.03322813  82.03952972    0           1       0       0
2321 355733.050   31.1207227273   121.2322291704   13.167    0.000    0.000    0.001  1.12300814  3.03327638  82.03934686    0           1       0       0
2321 355733.100   31.1207227274   121.2322291704   13.167    0.000    0.000    0.001  1.12300524  3.03328775  82.03912873    0           1       0       0
2321 355733.150   31.1207227275   121.2322291704   13.167    0.000    0.000    0.001  1.12298596  3.03334358  82.03892706    0           1       0       0
2321 355733.200   31.1207227276   121.2322291704   13.167    0.000    0.000    0.001  1.12301383  3.03340415  82.03879116    0           1       0       0
2321 355733.250   31.1207227277   121.2322291705   13.167    0.000    0.000    0.001  1.12305929  3.03346993  82.03862089    0           1       0       0
2321 355733.300   31.1207227278   121.2322291705   13.167    0.000    0.000    0.001  1.12307144  3.03350299  82.03846269    0           1       0       0
2321 355733.350   31.1207227279   121.2322291705   13.167    0.000    0.000    0.001  1.12305219  3.03353357  82.03827721    0           1       0       0
2321 355733.400   31.1207227280   121.2322291705   13.167    0.000    0.000    0.001  1.12305628  3.03361020  82.03808847    0           1       0       0
2321 355733.450   31.1207227281   121.2322291706   13.167    0.000    0.000    0.001  1.12306084  3.03359002  82.03792364    0           1       0       0
2321 355733.500   31.1207227282   121.2322291706   13.167    0.000   -0.000    0.000  1.12310250  3.03345299  82.03746830    0           1       0       0
2321 355733.550   31.1207227283   121.2322291706   13.167    0.000    0.000    0.001  1.12305435  3.03363438  82.03740573    0           1       0       0
2321 355733.600   31.1207227284   121.2322291706   13.167    0.000    0.000    0.001  1.12301251  3.03361948  82.03719991    0           1       0       0
2321 355733.650   31.1207227284   121.2322291706   13.167    0.000   -0.000    0.000  1.12302369  3.03369956  82.03701028    0           1       0       0
2321 355733.700   31.1207227285   121.2322291706   13.167    0.000    0.000    0.000  1.12300332  3.03378671  82.03689535    0           1       0       0
2321 355733.750   31.1207227286   121.2322291706   13.167    0.000    0.000    0.000  1.12303074  3.03377119  82.03682155    0           1       0       0
2321 355733.800   31.1207227286   121.2322291706   13.167    0.000    0.000    0.000  1.12294518  3.03378758  82.03668630    0           1       0       0
2321 355733.850   31.1207227287   121.2322291707   13.167    0.000    0.000    0.000  1.12292384  3.03380280  82.03656071    0           1       0       0
2321 355733.900   31.1207227288   121.2322291707   13.167    0.000    0.000    0.000  1.12288093  3.03380867  82.03644599    0           1       0       0
2321 355733.950   31.1207227288   121.2322291707   13.167    0.000    0.000    0.000  1.12289946  3.03387899  82.03628831    0           1       0       0
2321 355734.000   31.1207227289   121.2322291707   13.167    0.000    0.000    0.000  1.12290256  3.03385491  82.03616729    0           1       0       0
2321 355734.050   31.1207227290   121.2322291707   13.167    0.000    0.000    0.000  1.12288634  3.03389306  82.03603785    0           1       0       0
2321 355734.100   31.1207227290   121.2322291707   13.167    0.000    0.000    0.000  1.12286209  3.03391372  82.03593317    0           1       0       0
2321 355734.150   31.1207227291   121.2322291707   13.167    0.000   -0.000    0.000  1.12285834  3.03394534  82.03582074    0           1       0       0
2321 355734.200   31.1207227291   121.2322291707   13.167    0.000   -0.000    0.000  1.12285646  3.03389566  82.03569429    0           1       0       0
2321 355734.250   31.1207227292   121.2322291707   13.167    0.000   -0.000    0.000  1.12286036  3.03386556  82.03556929    0           1       0       0
2321 355734.300   31.1207227292   121.2322291707   13.167    0.000   -0.000    0.000  1.12289563  3.03388807  82.03545092    0           1       0       0
2321 355734.350   31.1207227293   121.2322291707   13.167    0.000   -0.000    0.000  1.12288504  3.03395900  82.03531661    0           1       0       0
2321 355734.400   31.1207227293   121.2322291707   13.167    0.000   -0.000    0.000  1.12289265  3.03398956  82.03517612    0           1       0       0
2321 355734.450   31.1207227294   121.2322291706   13.167    0.000   -0.000    0.000  1.12289040  3.03399179  82.03503224    0           1       0       0
2321 355734.500   31.1207227294   121.2322291706   13.167    0.000   -0.000    0.000  1.12290219  3.03402473  82.03488845    0           1       0       0
2321 355734.550   31.1207227295   121.2322291706   13.167    0.000   -0.000    0.000  1.12297087  3.03404794  82.03472987    0           1       0       0
```

Note: The file name extension can be "*.txt".

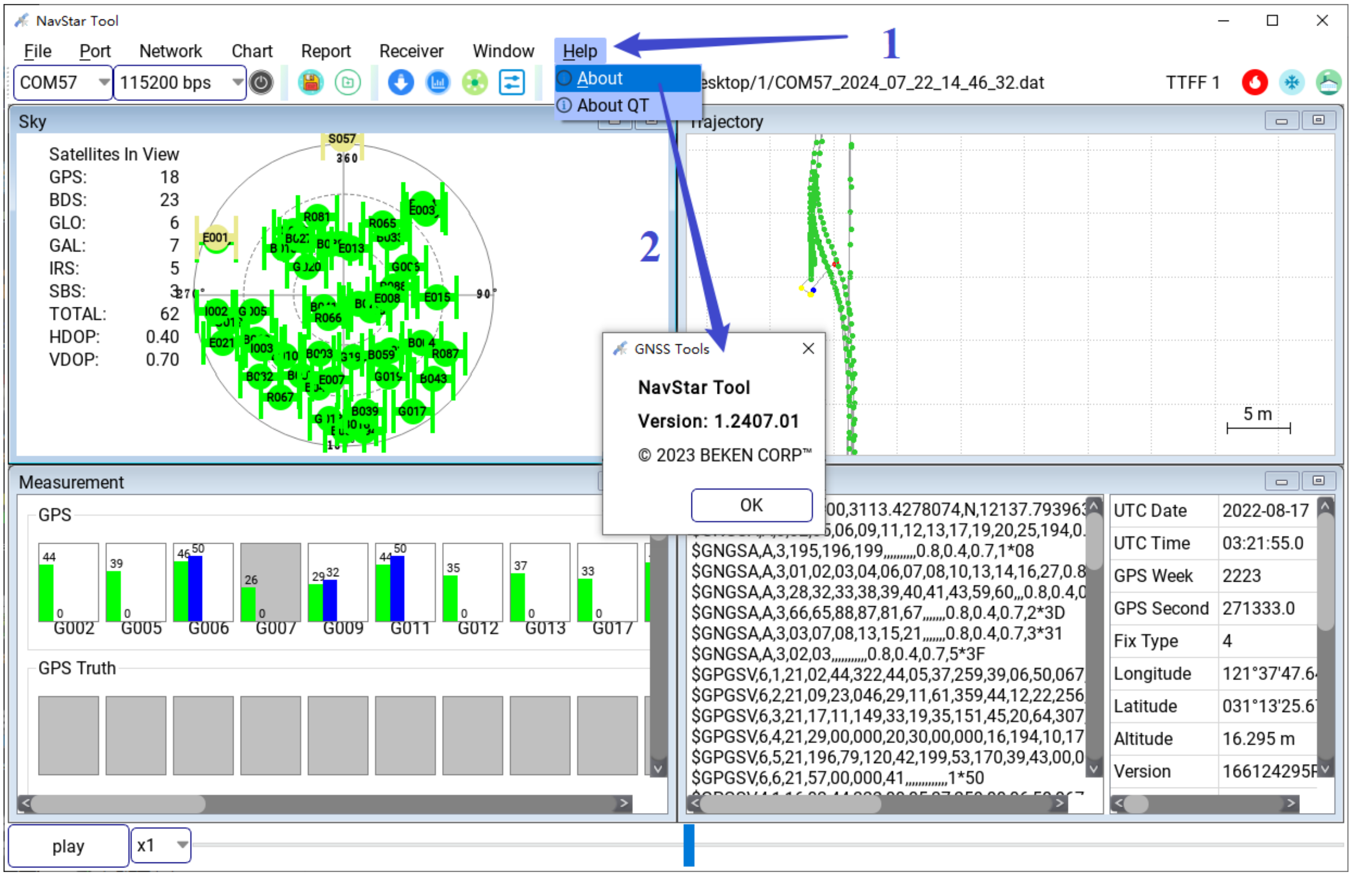
8.4 Dependent Environment Troubleshooter

Lost VC library files such as VCRUNTIME_XXX.dll

Please execute the VC library installer vc_redistXXX.exe in the corresponding version of the tool (x86/x64) directory to install the lost library files yourself.

8.5 Version Description

Click on the NavStar Tool menu bar [Help] -> [About] to view the tool version number information.



Revision History

Version	Date	Description
1.0	2022/1/20	Initial release.
1.1	2024/7/23	Added 8.1~8.3 to describe several file formats used by the tool; Basic operation added several commonly used test processes and methods; Add some details to the tool instructions.
1.2	2024/8/13	Added 3.3.4 Firmware Upgrade Exception Answer.

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